Program overview

19-May-2025 14:23

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Master Integrated Product Design			
1st year Master Integrated Product Design			
Faculty Core Courses			
IDEM4101	Delft Lectures on Design	5	
IDEM4201	Design Processes and Practices	5	
IPD Program Courses			
IDEM2100	Products Now Studio	10	
IDEM2101	Advanced Product Engineering	5	
IDEM2102	Cognitive and Psychological Foundations for Product Design	5	
IDEM2103	Empirical Design Research	5	
IPD Program Electives (minimum of 15 EC)			
IDEM2211	AI Products and Services	5	
IDEM2212	Intelligent Interactive Systems	5	
IDEM2213	Data-Centric Design for Connected Products	5	
IDEM2221	Strategic Integration of Sustainability in Design	5	
IDEM2222	Life Cycle Assessment Methods	5	
IDEM2223	Design Approaches for Sustainability	5	
IDEM2231	Material Driven Design	5	
IDEM2232	Computational Design	5	
IDEM2233	Fundamentals of Biodesign	5	
Design Studio Spring Semester (minimum of 10 EC)			
IDEM2200	Product Futures Studio	10	
2nd year Master Integrated Product Design			
Electives (minimum of 30 EC)			
IDEM101	Advanced Service Thinking and Design Practice	5	
IDEM102	Build Your Start-up 1: Idea Validation	5	
IDEM103	Build Your Start-up 2: Prototyping	5	
IDEM104	Creative Facilitation	5	
IDEM105	Exploring Design Intelligence	5	
IDEM201	Advanced Visualisation for Communication	5	
IDEM202	Advanced Visualisation for Design	5	
IDEM203	Biomechanics	5	
IDEM204	Child and Play Perspectives	5	
IDEM205	Cognitive Ergonomics for Complex Systems	5	
IDEM206	Contextmapping Skills	5	
IDEM208	Design in Health: Theory & Practice	5	
IDEM210	eHealth Design for a Healthy Society	5	
IDEM212	Food and Eating Design	5	
IDEM214	Lifestyle Research and Design	5	
IDEM215	Lighting Design	5	
IDEM216	PerForm the Unseen	5	
IDEM217	Strategic Automotive	5	
IDEM218	Systemic Automotive Design	5	
IDEM219	Videography	5	
IDEM220	When Images Remain	5	
IDEM221	Making and Prototyping Skills	5	
IDEM301	Animated Materials	5	
IDEM303	Computational Fabrication and MetaMaterials	5	
IDEM304	Design for Emerging Markets	5	
IDEM305	Designing Responsible AI	5	
IDEM307	Generative AI and Design	5	
IDEM308	Graduation Launchpad	5	
IDEM309	Repair & Recycling!	5	
IDEM401	BlueDot: Design Contest	3	
IDEM402	BlueDot: Organisation	5	
IDEM403	Research (3 ECTS)	3	
IDEM404	Research (5 ECTS)	5	
IPD Graduation Project (30 EC)			
ID4190-16	Graduation Project (IPD)	30	
IDEM2000	Graduation Project IPD	30	

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Master Integrated Product Design	
Introduction 4	A mandatory application for all courses (mandatory and electives courses) shall be made via the electronic applications system Osiris in the period that Osiris there to is opened. This period closes about 5 calendar weeks prior to the first day of the semester in which the programme starts; the faculty announces the exact deadline for application timely. The application for courses has to take place per semester, meaning for courses of 2 quarters at once. See http://www.io.tudelft.nl/osiris for deadlines and details.

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1st year Master Integrated Product Design

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Faculty Core Courses

IDEM4101	Delft Lectures on Design	5
Course Coordinator	Prof.dr. P.J. Stappers	
Course Coordinator	V. Tassinari	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This is a faculty core course for all IDE MSc students TU Delft is a world leading institution in design research. In this course, all first year Design Master students at TU Delft are given the chance to get in touch with the plethora of diverse and cutting edge design research present in the faculty, and further explore its potential for their own future career development. Delft Lectures on Design highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on design practice. The course comprises a series of interactive talks with researchers of the Delft Faculty of Industrial Design Engineering (and/or guest lecturers), who will be addressing how their research insights and expertise link to key contemporary positions in design discourse.	
Study Goals	Upon finishing this course, students are able to: 1. Analyze design research and specific topics in design research. 2. Formulate critical questions justified in the design research literature and other sources. 3. Demonstrate an understanding of the value of particular design research topics to society and the broader field.	
Education Method	The course consists of three cycles of three weeks each, during which students are first introduced by an IDE-researcher to a research theme and topics, followed by individual reflections as well as discussions in small groups. Based on these, each group formulates questions/provocations regarding a specific topic, which are then worked out to be communicated to the researcher in the form of a so-called provotype. A selection of these provotypes will be presented during a plenary event, followed by a moderated, interactive discussion between researcher(s) and students. At the end of the cycle each student individually reflects on the value of the provotype, the present and future relevance of the research as well as on personal lessons learned from the perspective of a future designer/researcher.	
Assessment	The study goals of this course will be assessed by: (pass/fail) Each cycle ends with an individual deliverable in which work of the three weeks, thus including results of the group work, is combined; deliverables C1, C2 and C3. C1 is assessed formative, where C2 and C3 are assessed summative with a pass/fail grading. Both C2 and C3 need to get a pass, to successfully complete the course.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dld-ide@tudelft.nl	

IDEM4201	Design Processes and Practices	5
Course Coordinator	Dr. P.A. Lloyd	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a faculty core course for all IDE MSc students. Design Processes and Practices introduces you to contemporary theory, methods, principles, and practices of designing in a broad and general sense. Exposing you to many diverse ideas about the design process - from theorists, scientists, methodologists, and practitioners - will allow you to appreciate the full range of factors that are in play when designing. In doing this, the course aims to help you develop a more nuanced vocabulary for talking about and doing the different activities that make up designing. Such a vocabulary will not only help you to understand the practice of design, but also help you to design your final Masters project.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the basic acts of design and which concepts, methods and approaches are appropriate. 2. Articulate theoretical aspects of design processes and practices. 3. Create a design process guide to support your future self in design projects.	
Education Method	The course features lectures, optional seminars, and a broad range of study materials (e.g. books, podcasts, videos, articles, blog posts, academic papers). As you progress through the course, you will individually produce a design process guide suitable to support you through your final graduation project. You will receive formative feedback at several points during the course. During the course, the different phases of the design process are referred to as acts, implying a narrative, dialogical and human structure taking place over a period of time and completed by different actors. Such a structure contains elements of prescription and description, as well as the latest developments regarding non-human actors. Course Structure: Week 1: Introduction and design guide sprint Week 2 & 3: Act one of design process Week 4 & 5: Act two of the design process Week 6 & 7: Act three of the design process Week 8 & 9: Act four of the design process Week 10: Course summary and conclusion	
Assessment	The study goals of this course will be assessed by: - Design Process Guide - Individual exam The final grade is pass/fail constituted by two summative individual assessments. Both assessments must be passed to pass the course.	
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Special Information	Contact: dpp-ide@tudelft.nl	

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

IPD Program Courses

IDEM2100	Products Now Studio	10
Course Coordinator	Dr.ir. S.F.J. Flipsen	
Contact Hours / Week x/x/x/x	8	
Education Period	1 2	
Start Education	1	
Course Contents	This is an IPD design studio. Developing designs that work! In the Products Now! design studio you will develop a design solution in the form of a product that fits a wider context. Join us for an ambitious design engineering challenge. You will dive into a highly complex client-based project using research, design, engineering, and team collaboration methods. Within a team of design-engineering students, you will design a product and service system. The team will start with an existing concept or prototype, and make a (re-)design which is circular and entails, besides a tangible prototype including a digital component. The process involves teams working on a highly-complex project, engaging an iterative design approach (ideate, build, test, evaluate), and solving intricate problems. Individually you will deep-dive in one of the identified challenges and come to solutions based on analysis and research. Together with your team you will further develop multiple prototypes combining the different challenges into a final demo model, which is presented to the client. This hands-on experience allows you to deepen and demonstrate your understanding of design and engineering throughout the design process. The course consists of the following phases: Introduction: Students form teams and analyse the design brief. Explore design opportunities: You will challenge your design brief through the following lenses: circularity, ergonomics, culture and context, service design, and engineering design. By making prototypes you will explore the design solution space and come to new concepts and designs, and their concerning challenges. Deep dive: In this phase you will individually analyse one of the identified main challenges through research using different methods. The outcomes will result in possibilities, requirements, constraints, and potential rebounds. Design: The different insights will be combined into a viable concept for embodiment. The teams subsequently further develop their product-service systems by means of iterative product engineering, and validate their findings by means of a high-fidelity demonstrator.	
Study Goals	Upon finishing this course, students are able to: 1. Analyse a complex (re)design challenge, using advanced methods and tools related to sustainability, engineering, and design. 2. Create designs following an iterative approach, and explore and validate potential solutions through engineering prototypes. 3. Develop an advanced and functional embodiment of a product-service system that integrates the sustainability, design, and engineering challenges generated throughout the project. 4. Document and present your research, engineering, and design outcomes for clients and the design engineering community. 5. Apply advanced techniques for design teams to maximise the team performance, and reflect on your individual contribution in the context of your design studio.	
Education Method	This course has a project-based learning (PBL) approach where design and engineering challenges needed for the embodiment are taught via available literature, online tutorials, and weekly key-note lectures and workshops. The student teams will be coached weekly on team-dynamics, progress of the project, and content. This is done by two coaches, one on design and one on engineering.	
Assessment	The study goals of this course will be assessed by (pass/fail): - Individual personal design journal: a report, which includes weekly progress, personal deep dive, and a personal reflection. - Group results: adequately documented in a report and, a final physical prototype, and a client presentation.	
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Special Information	Contact: pnds-ide@tudelft.nl	
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

IDEM2101	Advanced Product Engineering	5
Course Coordinator	Dr. J.H. Boyle	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This is an IPD programme core course. This course introduces advanced engineering techniques that can help you design better products. It addresses fundamental engineering principles and focuses on their practical application to the design and optimisation of products that are effective, efficient and sustainable. Students will learn how to use advanced software tools for modelling and simulation of important physical phenomena (solid mechanics, fluid mechanics, thermodynamics). They will also gain proficiency in writing custom Python code for analysis, simulation and data processing. Throughout the course, students will be taught to recognise the limitations of these techniques, apply them appropriately and critically evaluate their outputs.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the underlying principles and associated limitations of advanced engineering techniques, and contrast different techniques. 2. Apply advanced engineering techniques to a product or product component. 3. Analyse the results of advanced engineering techniques in order to evaluate the effectiveness, efficiency and sustainability of a design. 4. Evaluate the validity insights obtained from advanced engineering techniques.	
Education Method	Each week will consist of: A (two period) lecture to cover the relevant knowledge; a (two period) workshop to practise the practical application of the techniques; guided self-study (supported by a reading list and some pre-recorded video content) to supplement the lectures and workshops.	
Assessment	The study goals of this course will be assessed by: - Group Assignment, report based on practical work (40%) - Individual Exam (60%)	
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Special Information	Contact: ape-ide@tudelft.nl	

IDEM2102	Cognitive and Psychological Foundations for Product Design	5
Course Coordinator	Dr. rer. nat. C. Schneegass	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is an IPD programme core course. This advanced master course focuses on the foundations of cognitive ergonomics and psychology in and for product design. The course will cover the theoretical foundations of cognitive science and psychology, such as the information processing model, sensory perception foundations, memory theories, multitasking, and more, as well as practical techniques for designing user-centred products such as display or control design. For each theoretical topic block, the course will delve deeper into one or multiple of the following specialisations: (1) assessment methods, (2) cognitive reflection on design methods, or a (3) special interest case, inspired by current developments or news. Overall, students will develop a deep understanding of how people interact with products, both physical and digital, and how to design products that are usable and efficient to interact with.	
Study Goals	Upon finishing this course, students are able to: 1. Explain fundamental theories, key concepts, and latest research findings from cognitive ergonomics and psychology in the context of product design. 2. Critically evaluate existing product design solutions to ensure products, experiences, and interactions adhere to ergonomics and psychology principles. 3. Argue the appropriateness of metrics and methods for evaluating the usability and experience of products based on cognitive ergonomics and psychological theories.	
Education Method	This course adopts a blended learning approach, combining in-person lectures with online content for preparation and exercises. The weekly lectures provide a structured overview of the subject matter delivered by the core team and invited experts when experience in a central subject is not available in the core team. The lectures present information in a comprehensive manner while facilitating interactivity where possible. Guided self-study is promoted as an integral component and aligned with the weekly activities, allowing students to explore the material available online independently. These materials include curated resources and readings to prepare for and supplement the lectures, fostering a deeper engagement with and understanding of the course content. To allow students to individually assess their comprehension and learning progress, online quizzes and exercises (formative) are integrated into the curriculum. These interactive formats offer automated feedback (correctness) to gauge individual progress and reinforce theoretical knowledge. Quizzes can be repeated as often as the student wishes, and explanations for correct and incorrect answers are provided. The quizzes provide students with the possibility to prepare for the final exam. In case of questions, quizzes and exercises will be discussed in more detail in the lecture. By combining lectures, self-study, online quizzes and exercises, the course seeks to offer a well-rounded educational experience that accommodates diverse learning preferences and prepares students for academic and professional challenges.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4120-23 Technical Ergonomics and Product Experience.	
Assessment	The study goals of this course will be assessed by an individual written exam (100%) Summative Final Exam, 100% of final grade Formative: short weekly online quizzes and exercises (15 min estimated workload max) embedded into online lecture content as self-assessment and to maintenance rehearsal (formative). The quizzes and exercises will be a mix of different question-answer formats, including but not limited to multiple choice (with automated feedback) and open-ended questions (self-evaluation against answer example and/or rubric).	
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Special Information	Contact: cpf-ide@tudelft.nl	

IDEM2103	Empirical Design Research	5
Course Coordinator	Dr. H. Verma	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This is an IPD programme core course. This course provides students with a foundation in mixed methods design research and a framework of theoretical concepts and paradigms for rationalising, designing and conducting high quality empirical research, from instrumental research (for design) to scientific research (about/for/with design). It builds on and significantly extends the BSc Research for Design (IOB2-3) by introducing students to different epistemological paradigms such as empiricism, rationalism and perspectivism. The course employs an inquiry-based learning approach to introduce, practice and reflect on key research methods that encompass the diverse projects undertaken by IPD students during their design research projects. Case studies will be selected from the existing literature toprovide students with relevant assignments and motivation. Throughout the course, students will evaluate and discuss the appropriateness of various research methods in relation to the underlying research question, context, or objective. In addition, students learn to combine quantitative and qualitative approaches to meaningfully plan and conduct research to examine evidence, test hypotheses, or develop theory appropriate to a given context, design stage, and research question. Following are the key topics for this course: - Fundamentals of Empirical Design Research - Research Ethics and Data - Qualitative Research - Quantitative Research - Mixed Method Research	
Study Goals	Upon finishing this course, students are able to: 1. Explain the basic principles or theories behind empirical design research methods and scientific reasoning. 2. Evaluate the strengths and constraints of quantitative, qualitative, and mixed methods approaches to address research questions or design research outcomes. 3. Develop a mixed-methods research study protocol for a selected research question and justify the rationale. 4. Apply an appropriate analysis plan for your mixed-methods study and explain your decisions. 5. Review ethical considerations in empirical design research and discuss best practices.	
Education Method	The primary mode of instruction in this course will be weekly lectures. In addition, students will be provided with a curated reader (consisting of research articles and case studies) and bi-weekly formative assignments based on the suggested readings to supplement the classroom content. Students are expected to complete these assignments in a self-directed learning manner. Solutions to the assignments will be made available the following week. The hands-on sessions following the lectures organise and facilitate class-wide discussion of selected problems related to the lecture topic. These discussions are aimed at sensitising students to their bi-weekly assignment.	
Assessment	The study goals of this course will be assessed by an individual written exam (100%) Summative Assessment (100%): The final in-person, on-campus written examination will consist of a mixture of multiple choice and open-ended questions. The weekly assignments are designed to prepare students for the final exam. Formative Assessment: Students will be given bi-weekly assignments along with suggested readings (articles, case studies, etc.). Students will be expected to work on these assignments in a self directed manner. They will be provided with solutions to these assignments in the following week. In addition, weekly hands-on sessions after lectures, where instructors organise discussions on similar (but different) problems to the bi-weekly assignment, provide students with opportunities for formative feedback.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: ipdresearch-ide@tudelft.nl	

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

IPD Program Electives (minimum of 15 EC)

IDEM2211	AI Products and Services	5
Course Coordinator	Prof.dr.ir. A. Bozzon	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a programme elective of IPD theme 1: Digital Design and AI. This course is tailored for the Master's program in Integrated Product Design, and serves as an introduction to the theme Digital Design and Artificial Intelligence. The course provides students with a comprehensive understanding of the advanced digital technologies that support the functionalities of - and interaction with - AI-based products and services. It covers advanced Artificial Intelligence concepts, capabilities and (ethical and legal) principles, and contextualises the role of technologies (e.g. data, data analytics, Internet of Things) that enable the design and development of trustworthy and sustainable AI-enhanced solutions. Using a combination of Python scripting and rapid prototyping tools, students will learn how to develop and evaluate machine learning algorithms, language technologies, computer vision technologies, and human-AI interactions. Through a blend of theoretical knowledge, case studies, and hands-on activities, students will gain the skills needed to design innovative intelligent products and services. Such knowledge and skills will be put into practice during the Product Futures Design Studio.	
Study Goals	Upon finishing this course, students are able to: 1. Recall and describe key AI concepts and techniques. 2. Explain and analyse the capabilities and limitations of AI technology in the context of product and service design. 3. Design and prototype basic AI functionalities in realistic products or services. 4. Evaluate and critique the application of AI technology in products and services, focusing on their performance, ethics and sustainability.	
Education Method	The course uses a case study approach rooted in concrete examples of intelligent products and services. It combines required readings, interactive lectures, classroom discussions, tutorials, and hands-on activities focused on specific technology and design aspects.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5515 Advanced Machine Learning for Design.	
Assessment	The study goals of this course will be assessed by: - Individual Portfolio (30%): Every week, students will be presented with a formative exercise aimed at developing theoretical and/or practical knowledge. Each student will individually document their insights and learning in the form of posters, presentations, and/or demonstrations, which will be collected into a final portfolio that is handed-in and assessed at the end of the course. - Individual Exam (70%): a mix of multiple choice and open answer questions. Formative assessment for Individual Portfolio: Students will receive feedback on weekly formative exercises which gives them an opportunity to improve their deliverables before the hand-in of the individual portfolio at the end of the course. Formative assessment for final exam: Students will receive feedback on weekly on-line quizzes which prepares them for the final exam.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: aips-ide@tudelft.nl	

IDEM2212	Intelligent Interactive Systems	5
Course Coordinator	Dr. E. Niforatos	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 1: Digital Design and AI. This course provides students with a comprehensive understanding of Intelligent Interactive Systems (IIS) that leverage Large Language Models (LLMs) and Extended Reality (XR) to augment the perceptual and cognitive capacities of humans following the principles of Human-Computer Interaction (HCI). IIS refers to advanced technological solutions that combine AI with interactive user interfaces (UIs) to create dynamic, adaptive, and user-centric applications in actual or XR settings. Interacting with intelligent systems can be multimodal, multisensory and immersive, including touch-based, voice-based and text-based interactions with physical, visual, neural and virtual interfaces. Thoughtfully crafted intelligent interactive systems take into account the users perceptual and cognitive abilities and limitations, improve decision-making, increase productivity, facilitate knowledge sharing, enhance memory recall, prolong attention, and upgrade user experience (UX). In this course, students will demonstrate how Large Language Models (LLMs) and XR can enhance human abilities in challenging contexts such as healthcare, manufacturing, energy, creativity, and others.	
Study Goals	Upon finishing this course, students are able to: 1. Understand the principles and foundations of LLMs, XR, and HCI. 2. Demonstrate how LLMs and XR can be utilised to augment human abilities in challenging contexts. 3. Apply HCI design methodologies to meaningfully combine LLMs with XR.	
Education Method	This course will combine weekly lectures (2h) and hands-on (workshop) sessions (2h). Lectures and workshops will be grouped in Modules. Each module will be 1-3 weeks long and will conclude with a Group Report submission. Modules will include workshops that: - Familiarise the students with the technological aspects elaborated in the lectures by providing (code) examples, building blocks, and hardware demos relevant to the Group Report. - Support the students with the Group Report due at the end of a Module. The teaching team will help the students in groups with difficulties they may be experiencing in completing a Group Report. Group Reports will describe how students experimented with LLMs and XR for the context of their choice and in respect of the current module. Group reports will be reviewed based on a rubric. Feedback will be provided to the students as an opportunity to improve their work before they integrate it and submit it as final Group Portfolio (formative). Group Portfolios will be assessed at the end (summative). A set of knowledge quizzes will be provided throughout the course to enable the students to test their knowledge on IIS (formative). The quizzes aim to prepare the students for the final individual exam (summative).	
Assessment	The study goals of this course will be assessed by: - Group assignment (50%) - Individual exam (50%) Formative Assessment - Weekly Quizzes. Students will self-assess their knowledge based on a set of multiple choice knowledge questions. - Group Reports. Students will submit their group reports periodically receiving feedback on how to improve them. The collection of all (improved) group reports comprises the final Group Portfolio (see below). Summative Assessment - Individual Exam (50 %) assessing the comprehension of Intelligent Interactive Systems, their fundamentals, theories and principles. - Group Portfolio (50 %) consisted of a set of group reports graded at the end of the course. Grading Formula Final Grade = Group Portfolio * 0.5 + Final Exam * 0.5	
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Special Information	Contact: iis-ide@tudelft.nl	

IDEM2213	Data-Centric Design for Connected Products	5
Course Coordinator	J. Bourgeois	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 1: Digital Design and AI. This course will explore the intersection of data and design in the context of connected products. Students learn how to use data as a material for subjective inquiry, from thin to thick data and small to big data. They investigate what role data can play in the human-centred design process of connected and intelligent products and services. They explore ways to acquire, explore, interpret, reflect on personal data such as time series data from an activity tracker, text data from a products reviews, or 360 naturalistic videos from youtubers. They apply techniques to these data to enrich designers knowledge about the context and their empathy for the users. They use this material to connect with stakeholders, particularly people whose behaviour is represented in the data. Through a series of lectures and hands-on, group activities, students will develop the skills and knowledge needed to enrich the design process of their connected products through a blend of observational, telemetric and subjective data.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the role of data in human-centred design processes of connected products. 2. Apply time series, text, image and video processing techniques to support contextual inquiry. 3. Apply descriptive, explorative and participatory analyses to extract contextually rich insights from data. 4. Reflect critically and draw conclusion on the (responsible) use of data in the design for connected products.	
Education Method	The course is composed of a series of lectures and hands-on practice with extensive time dedicated to applying data analyses and processing techniques to understand the problem and solutions spaces. Throughout the course, students grow their understanding of the context, accumulating design-relevant knowledge from data which they deliver to their collaborators in the form of a short report and a Q&A session. Throughout the course, students have access to the open source Data-Centric Design Textbook and an open source series of short videos demonstrating tools and techniques. This material will be accessible for other students and general public beyond the Faculty. The data studio is open during the active time of the course to support students. Students receive formative feedback from teachers during workshop sessions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4130-23 Knowledge and Intelligence Design.	
Assessment	The study goals of this course will be assessed by: - Individual digital exam (60%), closed book, multiple choice and open questions - Group report with oral Q&A session (40%) Students compile their activities into a short report in which they describe the techniques applied, data use, results obtained and a reflection on insights. The assessment team use this to give the group a grade. During the oral Q&A session, the assessment team plays the role of a company executive committee and asks questions to check ownership and understanding of the work. Students will fill out a BuddyCheck questionnaire at the end of each module (four times in total) to monitor the healthiness of the group. Both the Q&A session and BuddyCheck can inform individual adjustment of the group grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dcd-ide@tudelft.nl	

IDEM2221	Strategic Integration of Sustainability in Design	5
Course Coordinator	Prof.dr.ir. J.C. Diehl	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a programme elective of IPD theme 2: Sustainable Design. Strategic Integration of Sustainability in Design will introduce students to the strategic business and design considerations of Design for Sustainability in practice. The first part of the course focuses on the strategic reasoning of how and why various organisations choose for sustainable innovations. The second part of the course will introduce the students to Sustainable Business Models (SBM) and Sustainable Product Service Systems (SPSS) and how to integrate them into design practice.	
Study Goals	Upon finishing this course, students are able to: 1. Understand the principles of SPSS and SBM models and their significance in the context of design and business. 2. Evaluate and debate how business organisations have integrated sustainability in a sustainable manner. 3. Apply SPSS and SBM models on two real-world cases with the same sector. 4. Analyse and compare business cases based on SPSS and SBM models. 5. Develop visual and verbal communication skills on the strategic integration of sustainability in design and business.	
Education Method	The course will offer interactive lectures, guest lectures and short movies. Lectures and short movies will focus on understanding how SBM & SPSS tools methods work, while guest lectures illustrate how these methods are used and integrated in real-world industrial design contexts. The lectures will be followed up by group discussions facilitated by the course team. Students will work in teams on two case study assignments. In between the two assignments, the group composition will be changed. During the first assignment, based on a given Harvard Business School case description, students will learn to understand and evaluate how large companies do integrate Sustainability in a strategic manner. During the second assignment, students will identify and select two sustainable businesscases within a similar industrial sector. They learn how to analyse and compare these cases by applying SPSS and SBM methods and tools. The student teams will have to do 3 times presentations with emphasis on different communication skills (oral, visual and debating skills). There will be weekly coaching sessions before and after the lectures.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4185-23 Strategic and Sustainable Design.	
Assessment	The study goals of this course will be assessed by: - Group assignment 1 (40%) - Group assignment 2 (50%) - Presentations (10%) Formative: - Groupwork, Coach feedback using a rubric. - Coach feedback on presentation skills (visual, oral and debating). Summative: - a final report of assignment 1 on the critical evaluation of the integration of sustainability in a business case (Harvard Business Case) (40%). - a final report of assignment 2 on applying and analysing SBM and SPSS tools in two cases in a similar sector (50%). - Visual and oral presentations of assignment 1 and 2 (10%).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: sisd-ide@tudelft.nl	

IDEM2222	Life Cycle Assessment Methods	5
Course Coordinator	B. Sprecher	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 2: Sustainable Design. Many students will be designing products that aim to be more sustainable than existing products. More generally, students are exposed to many claims of one product or material being more sustainable than another. These claims are based on a quantification of sustainability (for example the CO2 emissions associated with one unit of a product). Understanding how this quantification actually works is key if you want to make well informed decisions. In this course you will learn about the various quantitative assessment methods that are commonly in use. You will learn how to perform relatively basic Life Cycle Assessments (LCAs), and how to read and evaluate more comprehensive LCA reports. In addition to LCA, we will look at other methods such as the circularity calculators used in business context, and Mass Flow Analysis (MFA) and Input-Output Analysis (IOA) that are common in policy making and scientific context.	
Study Goals	Upon finishing this course, students are able to: 1. Perform a basic Life Cycle Assessment and communicate outcomes. 2. Evaluate comprehensive Life Cycle Assessment reports. 3. Analyse the relative strengths and weaknesses of LCAs, MFAs, IOAs, and circularity metrics.	
Education Method	The course will offer interactive lectures and guest lectures, and short workshops that focus on the use of LCA tools. Lectures will focus on understanding the how and why of quantitative environmental assessment. Guest lecturers will show how these methods are used in real-world context, for example in a design team or to make policy decisions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5314 Environmental and Circularity Assessment Methods.	
Assessment	The study goals of this course will be assessed by: - LCA assignment (50%) - Individual written exam (50%) There will be several formative assignments on making LCAs, and an individual graded LCA exercise. There will also be one individual exam.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: lcam-ide@tudelft.nl	

IDEM2223	Design Approaches for Sustainability	5
Course Coordinator	Ir. I.C. de Pauw	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 2: Sustainable Design. Design Approaches for Sustainability allows students to explore and apply 4 state-of-the-art sustainable design approaches: Eco-design, Circular Design, Transition design, Biomimicry. Students will learn how the different approaches work and apply them to a case. They will analyse and evaluate the differences in the approaches and outcomes for sustainability. The final objective is for students to learn what approaches will lead to what type of outcomes, and to know when to apply which approach. We offer this elective because designers can use different strategies to develop sustainable products, but creating a truly beneficial product is a challenge.	
Study Goals	Upon finishing this course, students are able to: 1. Explain four sustainable design approaches and their key concepts: Eco-design, Circular Design, Transition design, Biomimicry. 2. Analyse and Evaluate the differences between the key concepts of the four approaches, such as limitations, the solution space they offer, and the specific qualities they bring. 3. Apply the approaches and the associated tools towards sustainable solutions. 4. Construct an extension and/or improvement to one of the design approaches.	
Education Method	The first part of the course, students will apply all 4 methods to the same case (group work). In the second part of the course, students can choose one of the approaches based on their preference. This approach will be applied in-depth on a new case (group assignment) and students will deepen/extend one approach (individual assignment). In the final weeks, there will be a group reflection session for each of the 4 approaches that was applied in-depth.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5356 Sustainable Design Strategies.	
Assessment	The study goals of this course will be assessed by: - Group poster presentation (60%), demonstrating the results of the design process and application. - Individual assignment (40%), demonstrating the students ability to critically apply, adapt, and reflect on the design approaches for sustainability. - Students can miss a maximum of 2 contact moments.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: das-ide@tudelft.nl	

IDEM2231	Material Driven Design	5
Course Coordinator	Dr. E. Karana	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	This is a programme elective of IPD theme 3: Materialising Futures. As materials become alive, responsive and adaptive, and acquire new functionalities and expressions, how should designers work with them? How could they better understand and convey the potentials of such emerging materials for the interest of the planet and society? The Material Driven Design (MDD) Course is a didactic program with a strong focus on understanding and designing with emerging materials characterised by intricate temporal qualities. This Course equips students with the foundational knowledge, tools, and techniques necessary for practising MDD, with a particular emphasis on the technical, temporal and experiential qualities of the material, while also uncovering their latent positive impact on sustainability. The theoretical content of the course will be organised under 5 modules. [1] MDD Method [2] Understanding Material-Process-(temporal)Form relations (how to articulate Material Making Process and Taxonomy) [3] Characterization of Material Temporality (how to capture, analyse, and anticipate change in materials) [4] Experiential Characterization of Materials (how to explore material-people relationships) [5] Vision in Material Driven Design (how to reflect on the purpose of the material, integrating sustainable thinking)	
Study Goals	Upon finishing this course, students are able to: 1. Evaluate and justify the selection of key principles and techniques to perform a MDD project. 2. Analyse and interpret material-process-form variables and their relationships. 3. Apply tools and techniques to plan experimental set-up(s) to characterise temporality in materials and interpret their results. 4. Apply tools and techniques to plan user studies for experiential characterization of materials and interpret their results. 5. Debate the design potentials and broader sustainability implications of the material, to formulate a vision.	
Education Method	The education method includes: - Theoretical lectures introducing main concepts, methods, tools, and techniques under MDD; - Workshops and demos; - Practical lab activities on specific material cases (including living materials, smart materials etc.); - Materials characterisation studies; - Self-study involving readings of scientific papers and supplementary materials.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4140-23 Material Driven Design or ID5413-19 Material-Driven Design .	
Assessment	The study goals of this course will be assessed by an individual written exam (100%) We guide students in achieving the learning objectives through Interim formative assessments in the form of discussion sessions and quizzes. The summative assessment will be in the form of an exam. Formative Assessment (100% group work): - Discussion/Quiz Session 1 will be conducted after the first three modules. - Discussion/Quiz Session 2 will be conducted after the last two modules. Summative Assessment (100% individual work): - Written exam: Theory of Materials Experience and Material Driven Design; including close (multiple-choice) and open questions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: mdd-ide@tudelft.nl	

IDEM2232	Computational Design	5
Course Coordinator	Dr. Y. Song	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 3: Materialising Futures. Computational design empowers designers to utilise computer algorithms to automate aspects of the design process. It facilitates the exploration of a wide array of design options and can incorporate optimization techniques, particularly useful for digital fabrication. To use it effectively, a comprehensive understanding of geometry, data, algorithms, and related concepts is essential. Starting with computational thinking, geometric modelling, and CAD techniques, this course helps students learn computational design methods using Rhino Grasshopper. The curriculum covers various topics, including parametric modelling, algorithmic design, generative design, data-driven design, and extended reality (XR)-based evaluation. Through these areas of study, students establish a strong foundation in computational design methods and develop the skills to create and assess innovative designs.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the in-depth fundamentals of computational thinking. 2. Analyse and compare different computational design approaches. 3. Create designs by applying computational design approaches and necessary (new) algorithms using computational design tools. . 4. Assess the results of computational design through a suitable evaluation method. 5. Reflect on and communicate the advantages and disadvantages of computational design in a variety of product design contexts.	
Education Method	Lectures + Workshops, Self study, Assignments Self-study with the developed Grasshopper tutorials follows a flipped classroom approach, ensuring a swift mastery of geometric design tools. Hybrid lectures and workshops emphasize problem-based learning, with lectures covering key knowledge points, and workshops focusing on practical skill refinement. Assignments encourage problem exploration, effective problem-solving development, and the practical application of acquired skills.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5459 Computational Design for Digital Fabrication.	
Assessment	The study goals of this course will be assessed by: - Individual exam (50%) - Group assignment (50%) Summative individual exam: This computer-based exam comprises two parts: a series of questions based on learning objectives, and Grasshopper implementation tasks where students complete various tasks within a set time, with multiple approaches encouraged. Students have access to computers and the internet during the exam. Summative group assignment: In this assignment, students are given a design task, e.g. a personalised design. The group collects and translates requirements into designs using computational approaches. (Virtual) Prototypes, created through advanced manufacturing methods like 3D printing or VR, are then evaluated by the target group. The deliverables include a concise presentation with a video showcasing the design process and/or results, a comprehensive report supported by Grasshopper based program(s)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: comd-ide@tudelft.nl	

IDEM2233	Fundamentals of Biodesign	5
Course Coordinator	Dr. J. Soares de Oliveira Martins	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This is a programme elective of IPD theme 3: Materialising Futures. The Fundamentals of Biodesign course is an advanced introduction to biodesign with an emphasis on (micro)biology, biotechnology, and biodesign methods. It is focused on the fundamentals of living organisms (such as fungi, algae, and bacteria) and bioprocesses, and equips students with basic bio lab technical and research skills to grow, maintain, and observe living organisms, and utilise them for design projects. The course will be organised in iterations between practical lab experiments, theoretical/ inspirational lectures, and workshops. Students will follow an initial Lab Training, where they will get acquainted with Lab Safety Protocols, good laboratory practices and the basics of microbiological techniques. The content of the course will be conveyed through lectures and workshops starting with (1) a general introduction to Biodesign, and (2) the basics of Microbiology, where students will get acquainted with the biodesign field and the biological principles of distinct organisms; (3) a general introduction to biotechnology and their tools and applications in biodesign; (4) biodesign specific lectures and workshops, exploring the biodesign potential of specific organisms commonly used in biodesign. At last, we will address (5) bioethics and iodesign to provide the designers with the necessary ethical knowledge to research and design with living organisms.	
Study Goals	Upon finishing this course, students are able to: 1. Apply the principles of Good Laboratory Practices in a biological laboratory environment. 2. Explain the biological principles of the distinct organisms applied in biodesign. 3. Discuss challenges and approaches for using living organisms in biodesign. 4. Explain biotechnological processes essential for biodesign. 5. Perform biodesign experiments with living organisms.	
Education Method	The education method includes: - Theoretical lectures in microbiology and biodesign; - Practical lab activities introduction to microbiology (biolab) - Workshops in biodesign (biolab) - Self-study involving readings of scientific papers and supplementary materials.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5315 Fundamentals of Biodesign.	
Assessment	The study goals of this course will be assessed by: - Individual written exam (80%) - Group assignment (20%) Summative assessments: Final Exam: Individual (80% final grade) Laboratory Practice Test/Report: Group (2 students) (20% final grade) Formative assessments: Quizzes/Exercices/Essay Cases during the lectures and/or study time; Groups (appr. 3-4 students)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: fbd-ide@tudelft.nl	

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

Design Studio Spring Semester (minimum of 10 EC)

IDEM2200	Product Futures Studio	10
Course Coordinator	Dr. H.L. McQuillan	
Contact Hours / Week x/x/x/x	8	
Education Period	3 4	
Start Education	3	
Course Contents	This is an IPD design studio. Designing the future of product design Product Futures is an advanced design studio that explores emerging practices and future trends in product design. This studio challenges students to envision, experiment and realise the products, materials and technologies of the future, innovating toward novel product experiences, and sustainable ways of designing, making and living. Students select and tackle a project from among three themes: - Design for Sustainability: projects in this theme focus on advanced design methods and strategies for sustainable design to develop innovative products and services. - Digital Design and AI: projects in this theme focus on advanced digital technologies and emerging design and development approaches for next-generation intelligent products and services. - Materialising Futures: projects in this theme focuses on the analysis, fabrication and use of novel materials, and/or computational design and fabrication technologies. Building on research at the forefront of the chosen theme and working hands-on with emerging technologies, students develop, evaluate and present inspiring design/research artefacts and prototypes. Throughout the course, students closely collaborate with researchers and experts from each theme to experiment and discover their own approach on the intersection of technology, design and research.	
Study Goals	Upon finishing this course, students are able to: 1. Formulate/propose a design research project that is coherent, significant and innovative. 2. Experiment/innovate coherently on design research outcomes. 3. Compose/construct design research outcomes that are coherent, significant and innovative. 4. Analyse design research outcomes clearly and intelligibly. 5. Evaluate design research outcomes clearly and intelligibly. 6. Explain design research outcomes clearly, intelligibly, and engagingly. 7. Visualise design research outcomes clearly, intelligibly, and engagingly. 8. Organise your design research process effectively. 9. Evaluate your personal competencies.	
Education Method	The studio starts with an initial 4-week open studio phase for orientation, exploration, preparation and planning. Guided by the course team, students will collaboratively analyse design projects, conduct initial research and acquire necessary skills. Afterwards students enter the main design/research phase which consists of 3 4-week design/research cycles which will conclude with formative assessments (week 8, 12 and 16) and a final summative assessments (week 20). Students will work collaboratively, yet will be assessed individually. Throughout the studio, support and guidance will be provided in the form of coaching, lectures, group discussion, skills/studio classes, and feedback sessions for formative assessments.	
Assessment	The study goals of this course will be assessed by an individual assignment pass/fail (100%) 3 formative (week 8, 12 and 16) and 1 summative assessment (week 20) as follows: - Formative assessment 1 (proposal): presentation (individual, week 8). - Formative assessment 2 (midterm): Presentation, draft written submission + peer review (individual, week 12). - Formative assessment 3 (mock greenlight): design/research artefact/s, presentation, draft written submission + peer review (individual, week 16). - Final summative assessment: Design/research artefact/s, presentation, written submission pictorial format, self assessment. (individual, week 20).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: pfds-ide@tudelft.nl	

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

2nd year Master Integrated Product Design

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

Electives (minimum of 30 EC)

IDEM101	Advanced Service Thinking and Design Practice	5
Course Coordinator	Dr. H.M.J.J. Snelders	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The last decades have seen the application of already existing design practices to various service industries, with some degree of success. In this elective, we will explore how design can advance further in the service sector through a more specialised mindset and skill set for design research, ideation, and prototyping. The course is composed of two main parts: Part 1, on service design research, highlights how service users are always strongly connected to the production of a service. Special attention can be paid to this in design research, and we will offer hands-on exercises in methodologies that help you do so (notably Service Safaris and Journey Mapping). Part 2, on new service ideation and prototyping, highlights how value creation in services relies on collaborations between service users and producers that can only be partially controlled and planned for. Here as well we will offer hands-on exercises, using specialised methods (notably Bodystorming). Furthermore, the elective will work with real life cases from multiple service sectors (e.g. health, mobility, information, legal). Most exercises are group work, set up as a codesign effort where case interests will be weighed against your personal and group learning goals for developing your mindset and skill set for service design.	
Study Goals	Upon finishing this course, students are able to: 1. produce visual documentation of experiencing a current service that is creative and insightful, by using a specific service design tool. 2. Analyse and synthesise visual documentation of a current service that is creative and insightful of case information, by using a specific service design tool. 3. Co-create new service propositions that are relevant within the case, by using specific service design tools. 4. Demonstrate how the results from service design ideation and prototyping stem from a balanced dialogue between stakeholder interests and personal and group learning goals for competency development in service design.	
Education Method	Two types of learning activities are organised: 1) Through lectures, reading, and workshops students derive personal (yet theory-informed) learning goals for competence development in service design. These activities will take place in the first two weeks and middle of the course 2) Through real, case-based exercises with service design methods students can apply their learning goals in co-creation with case stakeholders. Next to a small individual first week exercise, these activities will take place from week 3 onwards.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5666 Service Design.	
Assessment	The study goals of this course will be assessed by: - Group work (80%) - Individual assignment (20%) All learning objectives will be assessed through formative feedback (throughout the course), and a summative review and grade (in week 4 and week 9/10).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: serdes-ide@tudelft.nl	

IDEM102	Build Your Start-up 1: Idea Validation	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 1 focusses on idea development and validation with actual customers, industry experts and other external stakeholders by applying methods such as problem interviewing, agile design & lean startup principles to assess desirability and first hints of feasibility and viability. Key question: is the venture idea promising enough to operationalize it in Q2?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment using generated evidence 2. Generate evidence by means of experiments and interviews with potential customers 3. Identify and make explicit the risky assumptions and weaknesses in their startup in a validation board 4. Reflect on their actions as an entrepreneur by writing an individual reflection paper 5. Explain the potential value of their startup to potential investors by crafting and doing an investor pitch	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. Well help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-1 Build Your Startup 1: Idea Validation.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (individual assessment) (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

IDEM103	Build Your Start-up 2: Prototyping	5
Course Coordinator	Ir. M.J.J. Buijs	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	The BYS 1 + 2 elective teaches how to build and validate the fundamental building blocks of a startup. Students start with a venture idea that slowly gets morphed into an actual startup with paying customers. Its one of the few, if not the only, course where students actually get to implement their own solution in the market. BYS 2 focusses on operationalizing the venture idea by forecasting and experimenting to validate: Market need. Getting actually first paying customers, Go2Market stategy, marketing campaigns Technology. Creating MVPs Operations. Start operating as a real business, with actual roles and goals, setting up the supply chain Setting up prices. Key question: is there initial product-market fit that proves the venture could become a success?	
Study Goals	Upon finishing this course, students are able to: 1. Justify their value proposition and initial customer segment by (mockup or real) sales. 2. Define a project plan for further operalisation of the startup after graduation. 3. Design & conduct experiments to research desirability, feasibility and viability of the venture. 4. Reflect on their progress as an entrepreneur & willingness to continue the journey & personal plan of action for the future by writing an individual reflection paper. 5. Show the progress of the startup & prove you are the team to do it to potential investors by crafting and doing an investor pitch.	
Education Method	In BYS courses, the emphasis is put on individual team coaching sessions, peer feedback and events where experienced entrepreneurs and experts (often course alumni) guide students pro-bono to validate their assumptions and build their idea. A mix of workshops and lectures is aimed to inspire and guide the students and forms a string of building blocks towards validating (BYS1) and operationalizing (BYS2) the venture idea. BYS courses are not ABOUT entrepreneurship. BYS courses are about DOING entrepreneurship. At BYS we tailor the journey around what your team needs at the very moment. Well help students to Build Your Startup, one step at a time.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5659-2 Build Your Startup 2: Prototyping.	
Assessment	The study goals of this course will be assessed by: - A Pitch of the startup, plan for further operationalisation and validation evidence (team assessment) (pass/fail) - A personal reflection using the EnJournal method (pass/fail)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bys-ide@tudelft.nl	

IDEM104	Creative Facilitation	5
Course Coordinator	K.G. Heijne	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	The Creative Facilitation course offers an in-depth experiential learning laboratory in the use of creativity techniques and the facilitation of creative processes. Facilitation in this context relates to someone whose role it is to work with a diverse group of people to ensure that a creative journey/process leads to innovative and implementable results with a high degree of consensus. The course starts with building a shared science-based vocabulary through literature and guided exercises and then moves on to hands-on experience through experiments and cases, each followed by presentation of results and reflection, in other words: experiential learning and reflective practice. During the course, we will look at the creative person, explore a wide range of creativity principles and techniques and develop methods to design processes for innovation teams. Creativity is not an end in itself, it is always applied to something; therefore, we will invite real clients with real problems: ranging from design challenges to new business opportunities to problem solving.	
Study Goals	By the end of the course, students will have an extended experience of organizing, facilitating and participating in creative processes. The actual outcome will vary for each student and depends largely on course objectives as set by each individual student. The course is set up in such a way as to provoke a reflective conversation on the creative process in small groups on the basis of theory and hands-on experience. Upon finishing this course, students are able to: 1. Apply the basic principles of facilitating creativity in group sessions. (Apply) 2. Compose a session plan and select appropriate creativity techniques. (Create) 3. Appraise the quality of creative sessions, by recognizing strong points and points for improvement in creative sessions the student attends either as facilitator or resource group member. (Evaluate) 4. Communicate clearly and interact professionally with the problem owner prior, during and after the creative session. (Apply)	
Education Method	Week 1-6: The first 6 weeks of the course we will apply the flipped classroom concept. The students will read theory and watch videos on the basic principles of facilitating creativity in groups. They will also make assignments individually. During the contact hours, the focus will be on guided exercises which gradually grow in complexity and scope. This is followed by sessions organized by students, guiding a group of their peer students from the course. In these sessions the students will experience a wide array of techniques and gain various insights in facilitation, group dynamics, problem analysis, idea generation and selection, and implementation. Each experiment is followed by an extensive (peer) reflection. Week 7-10 will be based on intensive experiential learning: setting up and leading sessions on all kinds of issues, ranging from design challenges to new business opportunities to problem solving. These sessions involve real issues from real clients. Each experience is followed by an extensive reflective conversation. The students will be encouraged to keep a journal to capture all learnings. This journal should also help them to get quicker into the right mindset necessary for this course. NOTE: Due to the steep learning curve, the hands-on approach and the group work, you will gain the best learning experience if you attend ALL days.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5325 Creative Facilitation.	
Assessment	This course was designed to focus on learning, understanding, and mastering the skills for facilitating creativity in groups, inspired by Beghetto (2005). To create optimal creative press, any creative output in this course will be assessed formatively only. The summative assessments are focusing on whether students will take the chance to experiment, how well they understand the principles and vocabulary, and how they capture what they have learned from their own and their peers experiments. Formative assignments on which the students will share and receive peer evaluation are: Learning contract (individual): an artefact, representing your personal learning objectives for this course and your commitment to achieve this. Practice sessions and creativity techniques (in duos): various rounds of sessions will be facilitated by students, on which they will reflect together. Draft session plan (in duos): before submitting the summative deliverable, students will get the chance to share and receive peer feedback on their session plan. The summative assignment (pass/fail) of this course is: Organize and facilitate a creative session for a real client (individual). This assignment will consist of 3 deliverables: Session Plan + substantiation of 2 activities (individual): to demonstrate that you can compile a session plan and know how to select relevant techniques. 2-pager Reflection (individual): demonstrating how well you reflect and integrate your knowledge and experience regarding Creative Facilitation, using the right principles and vocabulary. Client Report (individual): which is about properly and professionally communicating the session output to your client.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cf-ide@tudelft.nl	

IDEM105	Exploring Design Intelligence	5
Course Coordinator	Dr. R.S.K. Chandrasegaran	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Language is the primary means of sensemaking and communication among designers, but is often overlooked. With recent developments in artificial intelligence (AI), language has become a dominant way in which designers co-perform with AI models such as Large Language Models (LLMs). Understanding language is thus crucial to develop future design practice. Exploring Design Intelligence (EDI) will provide you with a theoretical and practical foundation in the tools and methods with which to analyse design language and conversation. You will use this foundation as the basis for engaging with AI systems to model and generate design conversation, and produce AI design methods that can enhance design thinking capability. The course will help you develop technical and research competencies alongside deeper thinking about the process of design.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the features of design conversation. 2. Analyse design conversation using existing theories of designing and psycholinguistic tools and methods. 3. Create an AI design method or tool using text generation techniques	
Education Method	EDI uses a blended learning model where lectures, videos, and reading material will be used to provide students with relevant theories and approaches. Students will individually explore the material and reinforce their understanding of concepts through exercises and explorations in the classroom. Assignments will help students learn to use the tools and methods introduced in the course. Depending on the number of students, coaches will be assigned to student groups.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5318 Exploring Design Intelligence.	
Assessment	The study goals of this course will be assessed by a report (100%) There will be one main assignment in addition to in-class activity. This will be a project involving the collection and analysis of design conversation data followed by the creation and application of an AI model to a particular design scenario. Formative Assessment: Using the above learning activities as a basis, students will investigate and verify their data using the provided reading material, the provided manual/computational tools and techniques, and their own literature review. Their progress will be assessed by the coaches through individual or group meetings, and feedback given to the students. Summative Assessment: The main outcome from the assignment is a report of their investigation along with the outcome of the investigation (e.g. a method, or approach).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: edi-ide@tudelft.nl	

IDEM201	Advanced Visualisation for Communication	5
Course Coordinator	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Expected prior knowledge	Level of visualization knowledge and skills should at least reach 3rd year TU IDE Bachelor visualisation courses.	
Course Contents	This course focuses on the accurate and communicative visualisation (depiction) of information and processes, which concerns (design) narratives such as journeys, roadmaps, data visualisation and design proposals, etc., aligned to pedagogic goals. The course offers a general course-wide and integrating module, relevant to the different tracks of which each concerns a different focal area. Theory module The theory module will include a theoretical grounding with the help of literature, deepening lectures, instructions and optional excursions. The integration of theory connects theory of visualisation with the teaching and applying of visualisation knowledge and skills. The tracks concern graphic design, process/system visualization, customer and patient journeys, and technology and visualization (subject to change). Tracks Various tracks offer different ways and application areas of visualisation, aligned to the interests of students from all IDE Masters programs. The tracks will concern both analogue and digital formats and tools. The course will utilise software, tools and learning platforms in order to facilitate both contents and management (communication) of the course.	
Study Goals	Upon finishing this course, students are able to: 1. Explain relevant knowledge and theories of visual representation, communication and (sequential) stories. 2. Clearly and convincingly create a visual depiction of information. 3. Apply both general and track-specific theoretical principles and methods that enhance the clarity and convincingness of the visualizations. 4. Apply both general and track-specific visualisation techniques and tools efficiently and vigorously. 5. Evaluate a visualization and formulate in words and/or visuals suitable ways of improving/developing the design visualization.	
Education Method	Practical education with theoretic grounding In-class active learning: a mix of instruction, guidance, individual work, continuous feedback, peer feedback, each student: 1 block of 4 hours in-class (including break). In order to support learning we will, in addition to the 4 hours in-class teaching, offer facilitated self study(FSS) (= optional support or guidance). FSS is unscheduled consultation, offered on a voluntary basis, both helpful for students who need extra support and for those who seek extra challenges.	
Assessment	The study goals of this course will be assessed by: - Visual individual report (80%) - Individual exam (20%) 1. Visual report containing portfolio of all exercises, works & visualizations produced during the course and includes considerations, decisions, reflection 2. Exam (visual and written) that concerns theory, and a substantiated and well-formulated evaluation of visual work.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: avc-ide@tudelft.nl	
Remarks	For this course a maximum of 100 participants applies.	
Material List	Depending on chosen course track: drawing tablets and Adobe software are required.	

IDEM202	Advanced Visualisation for Design	5
Course Coordinator	Ir. J.W. Hoftijzer	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Expected prior knowledge	Level of visualization knowledge and skills should at least reach 3rd year TU IDE Bachelor visualisation courses.	
Course Contents	This course advances the students knowledge of, and skills in visualization principles, methodologies and techniques, aligned to pedagogic goals. This course offers a variety of tracks to choose from. Besides bringing knowledge and skills to a higher level, the course also broadens the students knowledge of visualization and offers methodologies and techniques that are highly relevant for exploring and expressing design ideas and/ or concepts. It covers a wide range of theories and topics concerning sketching and visualization. Theory The course addresses the visualisation components that concern the exploration of thoughts (sketching as thinking), communication with stakeholders, imagination, and the promotion/ facilitation of collaboration, aligned to pedagogic goals. The theoretic element will be offered through literature, deepening lectures, instructions and optional excursions. The integration of theory connects theory of visualisation with the teaching and applying of visualisation knowledge and skills. Tracks The individual tracks concern different design areas, and will concern mostly generative practical exercises, active learning while guided by experts. Each track will have a clear relation with the specific field of application and specifc cases. Track topics concern e.g.: the human factor & interaction; visual creativity techniques; visualisation of mobility concepts; advanced design sketching (subject to change). Students will work with both analogue and digital tools and develop their understanding of generating visual depictions of complex objects, environments and people. Possible topics include: Theory: Spatiality, viewpoints and perspectives, rendering, light-dark (Chiaroscuro), Materials: properties and behaviour Colour theory Agency of sketching; methodology; Psychology of sketching; Psychology of perception; Skills: Development of motor skills, Advanced creativity- and ideation techniques; Visual exploration; The human factor; Human-product Interaction; Visualization for mobility (systems)	
Study Goals	Upon finishing this course, students are able to: 1. Explain relevant knowledge and theories of visualization as a thinking and development tool and language. 2. Create clear visual explorations of, and iterations on their concept sketch. 3. Apply both general and track-specific theoretical principles and methods that enhance the clarity and convincingness of the visualizations. 4. Apply both general and track-specific visualisation techniques and tools efficiently and vigorously. 5. Evaluate a visualization and formulate in words and/or visuals suitable ways of improving/developing the design visualization.	
Education Method	Practical education with theoretic grounding In-class active learning: a mix of instruction, guidance, individual work, continuous feedback, peer feedback, each student: 1 block of 4 hours in-class (including break).In order to support learning we will, in addition to the 4 hours in-class teaching, offer facilitated self study(FSS) (= optional support or guidance). FSS is unscheduled consultation, offered on a voluntary basis, both helpful for students who need extra support and for those who seek extra challenges.	
Assessment	The study goals of this course will be assessed by: - Visual individual report (80%) - Individual exam (20%) 1. Visual report containing portfolio of all drawings, sketches & visualizations produced during the course and includes considerations, decisions, reflection 2. Exam (visual and written) that concerns theory, and a substantiated and well-formulated	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: avd-ide@tudelft.nl	
Remarks	For this course a maximum of 100 participants applies.	
Material List	Depending on chosen course track: drawing tablets and Adobe software are required.	

IDEM203	Biomechanics	5
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Digital Human Models (DHM) are computer-based representations of humans that simulate interactions with virtual products or environments. Emerging in the 1960s, DHM has advanced significantly, especially with recent 3D scanning, AI, and computing innovations. It's a vibrant research field, vital for enhancing product function, fit, comfort, and safety in industries like automotive, aviation, defense, fashion, sports, and medicine. This course explores two key, intertwined DHM areas: 3D anthropometry and biomechanical modeling, particularly focusing on the musculoskeletal system (MSK). Initially, it provides an overview of human bones, muscles, and tendons, along with basic biomechanical principles. Students engage in workshops to learn about various DHM technologies, tools, and data. They will analyze dynamic human-product interactions, optimize them using DHMs, and set up physical studies to validate design choices. Example cases include designing sports equipment, bicycle seats, and respiratory masks. The course also covers tools, supports like chairs and beds, sports braces, wearables, and medical equipment.	
Study Goals	Upon finishing this course, students are able to: 1. Explain General Biomechanical Functions: Understand and articulate the roles of bones, muscles, ligaments, tendons, and joints in the human body, focusing on their biomechanical functions. 2. Apply Biomechanical Modeling Techniques: Use basic and advanced musculoskeletal modeling techniques for representing and analysing the biomechanics of human-product interactions. 3. Apply 3D anthropometric Modeling Techniques: Use state of the art methods, data, and tools for analysing fit in human-product interactions. 4. Propose Product Modifications to Optimise Human-Product Interactions: Examine human-product interactions, identify key influencing factors, and propose design modifications to enhance product function, fit, safety, and comfort. 5. Identify and Run Integrated Evaluation Approaches for Fit and Biomechanics: Develop and execute strategies for assessing fit and biomechanics in practical human-product interactions, selecting appropriate hardware, software, and criteria for effective evaluation.	
Education Method	Teaching through lectures, instructional videos, and hands-on workshops. Coaching of group assignments (4 students per group).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5301 Biomechanics.	
Assessment	The study goals of this course will be assessed by: Formative assessment through coaching and peer feedback. Summative assessment through two group assignments: (1) OpenSIM tutorials (40%) and (2) Developing a Wearable Sizing System (50% report + 10% presentation).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: biom-ide@tudelft.nl	

IDEM204	Child and Play Perspectives	5
Course Coordinator	Ir. M.A. Gielen	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	How do children (0-12 years old) perceive the world? How do they make sense of it through their play? How can designers cater for these play needs? These questions are at the core of the course on Child and Play Perspectives. The course is a mix of weekly lectures, playful exercises, design assignments and reflective mini-essays. In a group, you will set up, execute and analyse two generative sessions with English-speaking children visiting our faculty. You will derive themes for play from these sessions. You will study core theory and practices of childrens play and perform a series of play exercises yourself. And you will use all the insights you gain to ideate on playthings (toys in the broadest sense of meaning). Through this course, you will gain understanding and competence in designing responsible play products for and with children. This is a playful course. You are invited to play along!	
Study Goals	Upon finishing this course, students are able to: 1. Explain and apply effective techniques for generative user research and co-design with children. 2. Derive and communicate insights on childrens perspectives from data gained in generative user research and co-design sessions. 3. Explain key concepts of theory on childrens play and relate them to play experiences. 4. Create design ideas and argue how they build upon theories of play and support specific play experiences in children.	
Education Method	Self-study + group practicals + reflections + discussion The course is built up around a series of learning blocks, each containing material for self-study, a related practical exercise to perform, a reflection assignment plus a concise implementation assignment (compact design exercise). Each learning block concludes with an in-class discussion. Next to this, two sessions with primary school children will be held and reflected upon. All reflections will undergo formative assessment during the course and can be improved before the final summative assessment.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5184 Co-design and Research with Children or ID5185 Design for Children's Play.	
Assessment	The study goals of this course will be assessed by: Formative assessment of individual reflections and group design exercises of each learning block are offered on a continuous basis, depending on students chosen moment of sharing preliminary work. The rubric will be utilized as the basis for feedback on formative deliverables. Summative assessment: One single summative assessment takes place at the end of the course, of two items: 1) final version of the series of individual reflections (mini-essays). 2) annotated design map (the series of co-creation findings and compact designs); this is group work. PASS/FAIL assessment is applied in this course.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: chip-ide@tudelft.nl	
Remarks	Due to collaboration with a primary school, a maximum of 32 participants can take this course.	

IDEM205	Cognitive Ergonomics for Complex Systems	5
Course Coordinator	Dr. R. van Egmond	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Cognitive ergonomics is traditionally concerned with mental processes, such as perception, memory, reasoning, and motor response, as they affect interactions among humans and other elements of a system. Relevant topics include mental workload, decision-making, performance, human-computer interaction, and skill acquisition. Nowadays, products and systems are becoming more and more intelligent by the use of artificial intelligence. This requires a completely new understanding of the interaction that humans have with these systems or the systems with humans. We postulate that intelligent systems and humans have different views and behaviors of the same world and that this presents a new challenge to design to find the proper balance. These topics will be related to and examined in the context of complex systems. The course consists of two parts: 1. A theoretical part; 2. An empirical part. Theory will be exemplified by current and past research of our faculty. The empirical part will be based on a real-life situation of an encounter with a complex system and an abstraction made of this situation. This abstraction should mimic an aspect of a real-life situation and be tested in an experimental setting. The final step is to derive the implications for the real-life situation.	
Study Goals	Upon finishing this course, students are able to: 1. Identify potential cognitive ergonomics problems in users interaction with a real-world complex system. 2. Abstract an experimental protocol from a real-world problem and test it in a laboratory setting. 3. Communicate the research outcomes in a scientific manner using appropriate visualization methods.   4. Develop design recommendations for supporting users' cognition in the interaction with the complex system.	
Education Method	The course combines lectures on core topics, group project work, and coaching sessions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5332 Cognitive Ergonomics for Complex Systems 1: Understanding Complex Information Systems or ID5333 Cognitive Ergonomics for Complex Systems 2: Application into Research and Design.	
Assessment	The study goals of this course will be assessed by a group assignment (100%) Formative: Three status update group presentations on problem identification, study setup, and data analysis. Summative: One assessment consisting of two parts: (1) Complete project presentation and (2) annotated presentation slides.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cogergo-ide@tudelft.nl	

IDEM206	Contextmapping Skills	5
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Contextmapping is a design (research) method for including users and other stakeholders in a design process, being the expert of their experience. Underlying needs, motivations and dreams can be discovered by applying generative techniques in the research process. Over the past decade, the method, its principles and theories have been developed at TU Delft in close collaboration with Dutch and international researchers and practitioners. Contextmapping Skills goes in-depth in the fieldwork with coaching from expert practitioners at design agency Muzus and clients. Participation in Contextmapping Skills is open to all IDE MSc students who believe that end-users are important sources of information for leading new product development. DfI, SPD, IPD and PhD students.	
Study Goals	Upon finishing this course, students are able to: 1. Demonstrate and transfer empathic perspectives in both insights generation and insights delivery. 2. Design, conduct, and report an appropriate contextmapping study. 3. Analyse and organise data into relevant and meaningful insights. 4. Develop usable and impact-oriented deliverables. 5. Demonstrate open and constructive attitude and management.	
Education Method	In the course, teams of students perform a design research project for a client. The project follows the contextmapping process, at each step theory and instruction are provided in lectures, and literature is provided. At key moments presentations with feedback from peers, coaches, and experts take place. The course ends with presenting the findings to clients.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5663 Contextmapping Skills.	
Assessment	The study goals of this course will be assessed by: A final presentation and a report. Formative through weekly coaching and midterm assessment (report and presentation).	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: cms-ide@tudelft.nl	

IDEM208	Design in Health: Theory & Practice	5
Course Coordinator	Dr.ir. S.N. Paus-Buzink	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Healthcare can be regarded as a complex system in which organizations, healthcare professionals, and patients interact with each other. In addition, the healthcare system is shaped by many rules and regulations. Developing healthcare products or services can pose specific challenges, for example due to the complexity of healthcare systems, the diversity of stakeholders involved or regulatory requirements. Within this course, you will learn about the different domains in healthcare, apply different research techniques and design methods, various perspectives on design for health and what your role as a Medisign engineer can be. Topics covered include (amongst others): - Ecosystem and roles of stakeholders within healthcare systems - Examples of role(s) of design(ers) within healthcare - Regulatory requirements for developing medical devices - Risk analysis & designing for safety - (Design) research in the clinical setting - Ethics in medical (design) research - Health technology assessment This course is highly recommended for all students following the Medisign specialisation.	
Study Goals	Upon finishing this course, students are able to: 1. Navigate the broad landscape of design for healthcare. 2. Interpret relevant procedures, rules and regulations on ensuring quality and safety of products and services for healthcare. 3. Integrate of research methods and ethical practices from the medical domain into design research. 4. Propose a redesign of a product, service or product-service system by applying methods and theories provided in the course.	
Education Method	The course includes weekly lectures and workshops, in which TUD experts and professionals from different areas within the field of medical design share their perspective. In the workshops, you will work in a small group (2-3 students) on assignment(s) connected to the lectures, which compile the final report to be submitted at the end of the course. Understanding of provided literature and lecture content and application of theories and methods will be assessed in an exam at the end of the course.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5721 Capita Selecta Medisign or ID5752 Rules and Regulations for Designing Medical Devices.	
Assessment	The study goals of this course will be assessed by: - Final group report of the assignments (40%) - Individual exam (60%)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dhtp-ide@tudelft.nl	

IDEM210	eHealth Design for a Healthy Society	5
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	eHealth is often top-down technology-driven which has serious drawbacks on its uptake and negative effects on health disparity. This course takes a strict human-centered approach in the design of eHealth and has an overarching topic: 'How a human-centered approach of eHealth design can facilitate a healthy society' To this aim, the student will learn what eHealth is about, which specific design theories are relevant for it, and apply this knowledge in a practise specific case as well as in a general vision on the role of eHealth design for a healthy society. In this course students will be invited for the following health design journey in teams of 2: - Presentation of actual health innovation projects and design briefs by companies, external research groups, internal researchers. - Perform a critical analysis of interventions and create a preliminary future vision on the role of eHealth design for a healthy society. - Create design concept on design briefs - Present concept and vision to fellow students for peer feedback - Present vision and concept, including scenario, to stakeholders - Iterate vision and concept - Present report (including vision and concept) to tutors and stakeholders in final presentation. During the course interactive lectures will be provided on the following three perspectives of eHealth and the potential of design: - eHealth & healthy society (health philosophy, health-system and -value) - eHealth & health behavior (lifestyle, motivation) - eHealth & personalisation (groups (e.g. low SEP), profiling) The lectures will be given by faculty experts, guests, and PhDs/PDs. After each lecture, students will discuss with the expert the relevance of the topic to their health design project. Following these expert discussion, students will receive formative feedback on their project by the course coordinators. This course follows upon the elective Design in Health: Theory & Practice and is recommended for Medisign students.	
Study Goals	Upon finishing this course, students are able to: 1. Critically evaluate existing health interventions from a human-centered design perspective. 2. Apply the provided eHealth theory into the case-design by critically arguing for own design choices. 3. Create a convincing design concept based on eHealth theory for the given design brief by the client. 4. Create a vision on how eHealth design can contribute to a healthy society.	
Education Method	The students will work in pairs on a design brief that is formulated and introduced by a client. The course will have weekly meetings that involve a) presentations (by topic experts/ clients/ fellow students), b) discussions (with topic experts), c) formative feedback by individual group coaching by supervisors and/or clients.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5741 eHealth.	
Assessment	The study goals of this course will be assessed by a final report (100%) Each pair of students will be graded on their final report. The report will written according to a template that involves the following parts: - Evaluation of provided health intervention. - Embedding created (re-)design in provided theoretical topics. - Description of created design. - Related vision on the role of design for Healthcare. Grading of the report is done by: - Embedding the created design in the respective lectured topics by the respective topics experts. - Evaluation and vision by coordinators.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: ehealth-ide@tudelft.nl	

IDEM212	Food and Eating Design	5
Course Coordinator	Dr.ir. H.N.J. Schifferstein	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Even though designers are specifically trained to create and build new products, their contribution to food innovation is still relatively small. This is a pity, because providing people with good quality, healthy, and appetizing food products that are produced in a sustainable way, and that take into account local differences in consumption contexts, habits and rituals is quite a challenge. Furthermore, the regional, seasonal, and perishable character of foods challenges designers to find solutions that connect consumers with agriculture, trading, and processing methods. The food & eating design course makes design students aware of the different fields of knowledge that are relevant for designs in which foods play a major part. The emphasis of this course is on bringing design students in contact with the knowledge areas that are specific for food and are not part of their regular design curriculum. The course addresses: agricultural production, food technology, sensory perception and aesthetics, health issues, packaging, marketing, food culture, food systems, food industry.	
Study Goals	Upon finishing this course, students are able to: 1. Describe the context in which food product development takes place 2. Analyze the context for production, acquisition, preparation, consumption, and disposal of a food product. 3. Apply and reflect on the knowledge from the course topics within a specific food category. 4. Develop innovations addressing multiple challenges for a specific food category.	
Education Method	Lectures, guest lectures, workshops, coaching sessions. Every week 1 or 2 topics are treated. Students work in pairs on their writing assignments every week. The design challenges start in week 3. The design challenges are individual assignments. The start of the course is focused more on the theory and the writing assignments, whereas towards the end of the course students will have more time to work on their design challenges and will receive feedback from peers or coaches on their challenges in coaching sessions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5295 Food and Eating Design.	
Assessment	The study goals of this course will be assessed by: - Group report (50%) - Design challenge (50%) In weekly meetings, students, working in pairs, will give and receive formative feedback from peers and coaches based on the same criteria used later to evaluate the summative assignment. The final deliverable consists of a group report, which is a collection of the improved written assignments. (50% of final grade) In addition, the students work on a design challenge individually. This will be presented through a poster, presented orally in class. The students submit their poster, together with a short motivation and explanation of their idea as part of this assignment. The design challenge makes up 50% of the final grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: fed-ide@tudelft.nl	

IDEM214	Lifestyle Research and Design	5
Responsible Instructor	M. Filippi	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This course focuses on sociocultural research and form-giving. It delves into the concepts of personal and group identity, and the notion of 'lifestyle.' Throughout the course, you will conduct field research to explore the preferences, values, and cultural nuances of a chosen (presumed) lifestyle group. Insights gained from this research will be integrated into the design process, shaping a product, service, intervention, or a combination of these. The aim is to create designs that are not only functional but also appropriate and meaningful within specific contexts and for a particular group of people. You will use form-giving as a process to seek design-user resonance and to craft compelling narratives, with the concept of form including, but not limited to its traditional vision-oriented scope.	
Study Goals	Upon finishing this course, students are able to: 1. Investigate lifestyle-specific preferences/values/attitudes/activities/behaviours through desk and field research. (ANALYSE) 2. Argue the validity of the chosen lifestyle group based on research findings. (EVALUATE) 3. Design a product, service, or intervention tailored to the specific preferences/values/attitudes/activities/behaviours of a chosen lifestyle group. (CREATE) 4. Communicate your research and design outcomes in an effective, coherent, and appropriate manner. (CREATE)	
Education Method	The course combines weekly: - Lectures on theoretical knowledge. - Hands-on exercises. - Coaching sessions (in project groups of 4-5 students).	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5293-18 Lifestyle Research/Design.	
Assessment	The study goals of this course will be assessed by: Groupwork, and consists of two assignments: - Assignment 1 Group project (research), (60%). - Assignment 2 Group project (design), (40%). Assessment of assignments will be based on two deliverables: a group process report and a group presentation/exhibition. Mention of individual contribution to groupwork and personal reflections should be included in the report. A peer formative feedback session will take place approximately halfway through the course. The assessment will be based on the course rubric.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: lrd-ide@tudelft.nl	

IDEM215	Lighting Design	5
Course Coordinator	Prof.dr. S.C. Pont	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	With advancements in technologies, new opportunities are available for application in (public) spaces to enhance (smart) environments, e.g. (O)LED lighting and screens, daylighting systems, emerging photonic (bio-)materials. These generate new possibilities to apply lighting in our public spaces for: - Ergonomic (e.g. visibility, perceptual and cognitive fluency, safety), - health (influencing biorhythm, recovery in hospitals, daylighting for wellbeing), - mobility (city master plans, lighting for transport facilities), - sustainability (lighting design for darkness, bio-design), - aesthetical (attractiveness, beauty, light art), and - commercial (branding, atmosphere creation, persuasive environments) purposes, - influencing our environment visually and non-visually and thereby our affects, cognitions, behaviour and (mental) health. In this interdisciplinary studio we work with academics and professionals covering various fields from the hard technologies to soft perceptions, namely lighting designers from BeersNielsen, Atelier LEK, Koerner Design, Philips, ERCO, over a dozen of Dutch lighting designers and professors from the faculty of Architecture. The lecturing lighting designers also provide opportunities for asking advice, and in the final stage the lighting design café is organised to provide such professional advice which currently attracts about a dozen practising lighting designers. The course offers training in light and lighting, both theory and practice of lighting design, via lectures and hands-on practicals, workshops and excursions, plus the popular lighting design café. Throughout you will be working on a lighting design assignment going through a complete design cycle researching, (re)framing, and formgiving the intangible matter called light. For whom, what, where, and when it is needed.	
Study Goals	Upon finishing this course, students are able to: 1. Assess light and its impact on humans. 2. Compose an integrated lighting plan. 3. Elaborate the lighting plan technically and describe, visualise, present (and pitch) the final lighting plan. 4. Explain how to extend the design process beyond humans.	
Education Method	The course is based on a strong theoretical grounding, and applies the theoretical learnings to the design or research practice through application exercises, and skill development exercises, while integrating design, scientific knowledge, and professional skills into a practical group assignment.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5115 Lighting Design.	
Assessment	The study goals of this course will be assessed by: - Final Design presentation (50%) - Final report (50%) The quality and content of the group assignment is assessed in a summative manner. Formative assessments In the first part of the course students will get four individual homework assignments on which they receive written feedback via Brightspace. The course organisers are permanent coaches during the course; mandatory group coaching meetings will be scheduled on specific course afternoons to provide oral feedback and voluntary coaching moments may happen on appointment. Halfway the concept for the group assignment will be pitched after which feedback by peers and teachers will organized.	
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Special Information	Contact: ld-ide@tudelft.nl	
Remarks	For this course a maximum of 60 students applies.	

IDEM216	PerForm the Unseen	5
Course Coordinator	S.C.M. Brand-de Groot	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	In this course students will research unseen issues and render them tangible through (iterative) design processes in which formgiving the development, assessment, and determination of form takes centre stage. Designers can use their aesthetic sensibilities and communicative skills to support engagement with individual and collective concerns. Through their work, designers can spark and sustain public debate about these issues or revive controversies about long standing taboos. Matters of concern, moreover, of which many are intangible. Students will use an interdisciplinary set of skills to investigate how body language, individual perception and context influence the meaning of form in relation to design physical and/or digital artefacts. Students will investigate the communicative potential of form by means of embodied sketching and prototyping, experimental image and shape studies, observing and discussing. Students will develop their form studies into a design intervention that seeks to render tangible an unseen issue they chose to investigate. At the end of the course, they will evaluate the performance of their design intervention by means of a public presentation.	
Study Goals	Upon finishing this course, students are able to: 1. Analyse visual forms in their context using interdisciplinary skills like photographing, sketching and 3D modelling, artificial intelligence, discussing, philosophising. 2. Explore how they can use the human body as a ready yet rich source of information/inspiration for designing meaningful visions and designs. 3. Design with an intended meaning to make the intangible/invisible tangible, to bring about changes. 4.Communicate the vision of a design project by the conscious use of expression in form and performance. 5. Evaluate personal development, with reflections on insides and learnings.	
Education Method	Students discover the meaning of form by creating, observing, analysing, and discussing in groups. By using analogue (2D and 3D) and digital tools students will explore form by experience and doing. During the course we will be working with key skills like photographing, sketching, model making, artificial intelligence and bodily movements. In class we will do small individual and group assignments that we discuss in class. In the second half of the course, students will work in groups of 4 (or 5) students in a large design assignment including the performance at the end of the course.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5297 perFORM.	
Assessment	The study goals of this course will be assessed by: - Group assignment (40%) - Individual written exam (60%) During the course each week we will discuss with peers and teacher(s) to exchange experiences and outcomes. In an individual visual expression the student will perform the research, the design process, and the final outcomes/design of small and large assignments. The student must reflect and evaluate on the experiences, theory, and discussions in class. A final evaluation will be written. Next to the individual expression the students will be assessed on the group work: the design of a meaningful and coherent final form expression as an interdependent whole.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: perform-ide@tudelft.nl	

IDEM217	Strategic Automotive	5
Course Coordinator	E.D. van Grondelle	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This course aims to bridge between styling and strategy and make form giving strategic. The automotive industry is internally and externally undergoing unprecedented transformation. Designers must become strategic to embrace internal developments such as new propulsion technologies, new human product interaction, new ownership models and vastly increasing competition from new market entrants. Designers must also become systemic as car companies are transforming into ecosystems to adapt to the emerging holistic mobility paradigm which rebalances the design rationales for all mobility modalities. To accommodate this new design paradigm the automotive design process, which is traditionally rooted in tacit knowledge, requires a methodical framework. In this course we synthesise design methods and new styling management tools into a comprehensive set including tools and methods such as Form Strategy and Form Hierarchy, Buys, Porter and VR, Systemic Visualisations and the Metaverse. Note that while this electives case studies and exercises are rooted in the automotive domain, its learnings are equally applicable in other (mobility) domains.	
Study Goals	Upon finishing this course, students are able to: 1. Students can analyse a brands identity and image through a variety of secondary and primary sources. 2. Students can analyse products design (meaning of form) within the branding context, with respect to their underlying technology and business context. 3. Students can apply a variety of business models and methods to negotiate design with other disciplines and stakeholders and create a design strategy brief fit to automotive ecosystems. 4. Students can convincingly create a visual form language as the outcome of branding-by-design driven design brief.	
Education Method	Study load is spread evenly along the course, guided by case studies. This course consists of weekly 4-hour sessions, each of which combines - Short lectures (case studies), two or three per morning - Dialogue - Visualisation exercises - Project work - Impromptu presentations	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5249 Strategic Automotive.	
Assessment	The study goals of this course will be assessed by a final report (100%) Formative assessment by coaches and peers will be provided weekly in response to impromptu presentations. Summative assessment consists of one 3-member team assignment: a final presentations handout in the form of a report that includes processing of the feedback that was given upon the presentation. Students also write a personal evaluation, which may influence individual marks.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: straut-ide@tudelft.nl	

IDEM218	Systemic Automotive Design	5
Course Coordinator	E.D. van Grondelle	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Course Contents	This elective belongs to our facultys strategic research theme Mobility. Automotive Design is no longer solely about personal vehicle design. Automotive Design is about applying its specific knowledge to mobility design in its full width, to rebalance individual wants and collective needs. We synthesise this tacit automotive design knowledge with the predominantly rational design paradigm of public mobility. In this elective, you will unravel the complexity of future mobility and society, to understand what meaningful mobility solutions are. By analysing and visualising THE CONTEMPORARY MOBILITY SYSTEM, you will identify what is relevant in todays context and identify your own position in the holistic system i.e., act as a systemic designer. You will develop a sense of urgency to explore what is appropriate within a future context. For that future context, you design an ADVANCED FUTURE CONCEPT VEHICLE, which is derived from your long-term future vision. How will mobility develop, and how should it be experienced? What is your futures personal, public, or shared vehicles reason for being? From there, you must design an INTERMEDIATE VEHICLE that bridges contemporary mobility to the advanced future concept.	
Study Goals	Upon finishing this course, students are able to: 1. Students can identify and explain their own position and power in the mobility domain and take a stand (systemic design). 2. The student can analyse a future mobility context in relation to a constructed future society, reorganise and correlate its factors into a holistic perspective, and construct a future vision. 3. The student can apply a context driven design process, balance between individual wants and collective needs, and translate its outcome into design manifestation. 4. The student can apply a variety of design tools, including AI, and convince stakeholders at a visual level as is expected in the automotive design domain.	
Education Method	This course is guided by a context driven design process which requires participants to keep pace. Weekly half day (4 hours) sessions include - Short lectures - Class discussion - Short progress presentations - Project work with coaching	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5242 Automotive Design and Mobility.	
Assessment	The study goals of this course will be assessed by: - Individual assessment 1 (50%) - Individual assessment 2 (50%) Study load is spread evenly along the course, guided by the context driven design process. The coherence throughout between analyses, storytelling, and manifestation, is fundamental. Weekly coaching project criteria will be informally and through dialogue addressed in class. Two individual summative assessments: an annotated presentation during week 5 addresses the storytelling aspect of the design project, and a presentation plus final visual in week 10, which assesses the manifestation i.e. the design.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: sad-ide@tudelft.nl	

IDEM219	Videography	5
Responsible Instructor	Dr.ing. M.C. Rozendaal	
Contact Hours / Week x/x/x/x	4	
Education Period	2 4	
Start Education	2 4	
Course Contents	Video, with its inherent narrative structure, visual and auditory richness, and ability to visualise future contexts, offers much potential for design and designers. In this course you will unlock some of this potential by learning how to use videography as a tool in your design process. You will be taught videography techniques and cinematic principles, and apply these in hands-on assignments. Following a general introduction, you choose one of three different tracks to deepen your skills: 1. Documentary Videography offers designers a way to capture the richness and detail of existing contexts of use, by showing real people performing real actions in real situations, while expressing genuine feelings, concerns and emotions. 2. Animation The potential of animation to combine imagery, music and sound in interesting new ways, offers designers a powerful medium to explain abstract ideas, convey a rich pallet of emotions, and simplify complex subject matter. 3. Fiction Videography offers designers the possibility to envision future design ideas through scenarios that communicate their form and use within particular contexts of use, and hereby addressing their potential impact in convincing and critical ways.	
Study Goals	Upon finishing this course, students are able to: 1. Apply videography skills in different stages of a video production. 2. Create quality media products for different design purposes. 3. Evaluate the value of videography in the design process.	
Education Method	The course will combine lectures and workshops, coach meetings, and make use of self-study. We start with a general (3-week) track in which students learn basic videography skills and fundamentals. Then in the following 6 weeks, students choose one of three different tracks to deepen their skills; (1) documentary (2) animation, (3) fiction. Students are consulted about their preferences a week before the course starts. Based on this, a balanced division across tracks will be made.	
Assessment	The study goals of this course will be assessed by: - Group portfolio (80%): Students build a video-portfolio based on group-work that will be assessed at the end of the course. - Written evaluation (20%): Students write an individual evaluation at the end of the course by reflecting on the value of videography in the design process.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: video-ide@tudelft.nl	

IDEM220	When Images Remain	5
Course Coordinator	Dr. M.W.A. Wijntjes	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	The course, titled When Images Remain, acknowledges the contemporary visual-centric world. It recognizes the significance of images as carriers of extensive, often invisible, information and explores the evolving role of visuals in human-machine knowledge transfer. The course unfolds the histories of image-making on a local and global scale, considering images as windows to the past and future, reflecting societal changes, artificial intelligence, and evolving values. This course delves into visual design through three distinct perspectives. First, it engages in theoretical analysis of existing image collections (see our archive), incorporating insights from art history, design, perception, and computer science. The critical examination considers both the creator's (sender) and the audience's (receiver) viewpoints, introducing the Vision and Depiction framework, which merges formal elements with contextual factors. Second, the course integrates empirical investigations. Students learn to conduct visual experiments, analyze data, and visually represent findings. The third perspective explores how computers interpret images, emphasizing the contribution of computational analysis to visual research and paying attention to ways in which visual research findings are shared and communicated.	
Study Goals	Upon finishing this course, students are able to: 1. Apply formal analysis based on the Vision and Depiction framework to gain in-depth experience into the expressive possibilities of visual interaction parameters. 2. Evaluate visual research using visual paradigms and computational approaches. 3. Reflect on the differences between human and computational interpretations of images, as well as the importance of developing an understanding of 'what makes an image' 4. Visually communicate the research results in a manner based on the Vision and Depiction framework.	
Education Method	Weekly lectures (2 hours), 3 workshops (2 hours) and 3 peer review moments.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID4230 Visual Communication Design.	
Assessment	The study goals of this course will be assessed by an individual assignment (100%) All assignments are individual. Three formative assessments including the analysis and evaluation of images, and a reflection. The summative assignment will consist of the revised formative assessments, with the addition of a poster.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: wir-ide@tudelft.nl	

IDEM221	Making and Prototyping Skills	5
Course Coordinator	Dr.ir. T.J. Jakiewicz	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Course Contents	In the prototyping elective, students will select from a relatively open range of small modules that cover different prototyping skills. Students start the course with a reflection on their professional identity as a designer, and individually identify 4 modules from the provided prototyping skill-tree that would support them in strengthening that identity. During the course, they go through the module-specific online tutorials (8h), and then spend time applying that knowledge to a project of choice, reflecting on and documenting their process (8h). Through most of the course students work autonomously. Support staff hold consultation hours where students can come and ask questions and discuss possibilities and strategies. SAs provide technical support. At the end of the course, students work in small groups to create a new or improved module tutorial. Negotiation: At the start of the course, students formulate their prototyping-skill ambitions and project-related prototyping needs. Course staff supports students strategically in selecting the right modules and collaborates with students on improving and developing new modules. Articulation: Students are responsible in the course for a combination of: - Redeveloping/improving an existing module to bring it up to date or add in additional ways that the skill can be developed. - Designing and testing a new module that teaches a new skill - based on talking with an expert, they formalise that knowledge, and make it communicable for future students.	
Study Goals	Upon finishing this course, students are able to: 1. Autonomously expand their own prototyping skillsets. 2. Select among various prototyping options based on skills, needs, and context. 3. Determine appropriate prototyping skills required in a project and critically evaluate the prototyping process. 4. Communicate how to use particular prototyping skills to others in an engaging and professional way and are able to contribute to practical knowledge generation exchange in a professional prototyping community.	
Education Method	Self-directed work based on online material. Optional weekly consultation moments in which students can obtain feedback and support.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5231 Experimental Formgiving of Visual Information.	
Assessment	The study goals of this course will be assessed by: Receiving a pass/fail grade based on submitting a) individual portfolio of filled in prototype card forms, and b) new tutorial or tutorial improvement submitted as a group. Submission of complete required deliverables, demonstrating autonomous completion of tutorials and new group contribution, guarantees a pass. Pass/fail assessment is performed by the core team at the end of the course with assistance of SAs. Students provide peer-feedback on each others tutorial contributions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: maps-ide@tudelft.nl	

IDEM301	Animated Materials	5
Course Coordinator	Prof.dr.ir. K.M.B. Jansen	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	This course is an introduction to the field of Animated Materials - materials that change during fabrication and/or use time, their relevance and application in the development of products, and the use of tools for the design of temporal materials and products made with them. Animated Materials facilitate transitions towards multi-situated, long-lasting artefacts that may enrich society and improve environmental impacts. A general introduction in a range of animated materials will be given, however, a selected animated material context will be explored in detail (for example: shape-memory materials, multimorphic textiles, e-textiles).	
Study Goals	Upon finishing this course, students are able to: 1. Discuss basic material components and examples of Animated Materials in terms of their context and impact. 2. Apply design and prototyping methods for Animated Materials. 3. Evaluate your design proposals and prototypes to determine relevant approaches. 4. Experiment with design and prototyping methods for Animated Materials. 5. Develop at least one Animated Material prototype/demonstrator that speculates a future application.	
Education Method	After a very focused exploration of fundamental materials and structures, the theories, tools and methods for (selected) animated materials will be conveyed through lectures and complementary workshops, in which students will have a chance to practise the given theory and techniques within teams of two or three. These lectures and workshops will be highly inspirational supported by recent example cases from the material science and design domain, and guest lectures where possible. Parallel to this the students will perform a short literature study on the performance and affordances of different classes of physically animated materials. After this, the students will conduct an experimental design research project to apply the theory they learned and create material concepts and applications. Herein, hands-on material experiments in the Lab (e.g. material tinkering, DIY material making, construction, modification), and studio discussions will be the main activities. While most of these activities will require self-study time, the experts will be available for individual meetings. Tools to support the development of materials that change will be delivered (low value physical prototyping, modelling, CLO3D for example) Week 1: Introduction to course, state of the art of Animated Materials (Lectures and discussion; instructions for benchmark study) Week 2: Animated Materials 1 Introduction to fundamental material properties of selected animated material (Lecture, examples, demonstrations and workshop) Week 3: Animated Materials 2 Deepening knowledge of Animated material (lectures and workshops) Week 4: Open Studio/Coaching Week 5: Midterm presentation (Results of Benchmark study and proposed direction for remainder of project) Week 6: Open Studio/Coaching Week 7: Open Studio/Coaching Week 8: Open Studio/Coaching Week 9: Open Studio/Coaching Week 10: Exhibition of prototypes and report submission	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5419 Multimorphic Textiles.	
Assessment	The study goals of this course will be assessed by: - Individual assessment (20%) - Group work (80%) There will be four main deliverables in the course: [1: 20%, individual] Benchmark report; [2: 20%, group] Prototype; [3: 20%, group] Video of prototype in future application scenario; [4: 40%, group] Project report, describing the design research process; The course will end with an exhibition in which students will present their prototypes and videos	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: am-ide@tudelft.nl	

IDEM303	Computational Fabrication and MetaMaterials	5
Course Coordinator	Dr.ir. E.L. Doubrovski	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	In this course we introduce a new way of thinking about and working with materials in digital design and 3D printing. Through computational fabrication, we can tailor material behaviour and experience. This is in contrast with common practice prescribed by traditional manufacturing, where materials are given or chosen. With computational fabrication, we not just consider the constraints of 3D printing, but actually leverage the specificalities of the printing process to obtain structures with desired properties. To achieve this, a deep understanding of the inner workings of 3D printing is needed, as well as learning how the material responds to the 3D printing process. Blurring the boundary between 3D CAD and 3D printing, students will learn to use Grasshopper to define their designs as a set of machine instructions for the 3D printer. They will apply this to both Material Extrusion and Material Jetting techniques. Starting with simple structures, evaluating and testing their outcomes, and culminating in the application of their acquired knowledge to fabricate a prototype.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the differences between various multi-material printing processes 2. Apply a computational fabrication approach using Rhino Grasshopper 3. Analyse the effect of 3D printing parameters on the properties of a structure 4. Conceptualize and prototype a product in which the behaviour is achieved by leveraging the 3D printing process 5. Evaluate the 3D printed result, is the desired behaviour achieved?	
Education Method	Lectures, Workshops, Assignment	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5459 Computational Design for Digital Fabrication or ID5454 Digital Materials.	
Assessment	The study goals of this course will be assessed by: - Summative individual exam (50%). A computer-based exam, where students are asked to solve multiple tasks in Grasshopper. - Summative group assignment (50%). The groups are given a design task for which they devise a computational fabrication workflow, fabricate multiple prototypes for testing, and present the final design. The deliverables are a demo and written report.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: comfab-ide@tudelft.nl	

IDEM304	Design for Emerging Markets	5
Course Coordinator	Prof.dr.ir. J.C. Diehl	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	This course is focussed on developing innovative product (service) systems solutions for emerging markets, in specific for low-resource settings. People in these settings are often confronted by a range of Sustainable Development Goal (SDG) challenges in their daily life such as a lack of access to basic healthcare (SDG3), food (SDG2) and education (SDG4), poverty (SDG1), and inequality (SDG10). While solving these challenges specific constraints of low-resource settings such as a lack of connectivity, infrastructure, low levels of literacy and limited (household) budgets have to be taken into account. In order to develop solutions for these SDG related challenges in low-resource settings, new design approaches and design mind-sets are needed like for example frugal innovation or design for the other four billion. These include deep listening to local consumers, new business models (like pay as you go), affordable but smart technologies, and new product / service infrastructures. In the first place this course will introduce the students to new research and design approaches for these emerging markets along a range of cases like smart health diagnostics, distributed renewable energy systems, and clean cooking. The student will be as well introduced to and prepared for the internationalisation of the industrial design profession, such as, entering international markets, international product policies, globalisation and working in international teams. Lastly, the students will reflect and debate about what the role of (international) industrial designers is in these contexts, as well as how should be taking into account aspects such as design ethics and decolonization.	
Study Goals	By the end of the course students will have an extended experience in analysing a low-resource context in an emerging country, able to apply context-specific design methods, and generate initial ideas for interventions. During the course the students will be provoked to reflect on their role and position as a designer in the context of low-resource settings. Upon finishing this course, students are able to: 1. Analyze and describe specific use-contexts and eco-systems in emerging markets. 2. Apply design methods and tools specifically for designing for emerging markets and low-resources settings (i.e Design for Well-Being, Frugal Innovation, context specific design). 3. Create innovative ideas for product (-service) solutions for low-resource settings in emerging markets. 4. Evaluate what the role and responsibilities are of designers active in emerging markets from an ethical perspective.	
Education Method	The course includes weekly lectures in which TU Delft experts and professionals from different backgrounds (NGO, business, non-profit consultancy, and academics) share their perspectives on Design for Emerging Markets. The lectures will be followed up by reflective discussions moderated by the course staff. The students will work in teams of maximum 5 students on a real life case with clients in the Global South. The students will be in direct contact with the client and weekly coached by the course staff. This will result into a final presentation and team report at the end of the course. The course will be wrapped with an individual reflective exam with open questions.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5050-18 Design for Emerging Markets.	
Assessment	The study goals of this course will be assessed by: - Individual assignment (30%) - Group assignment (70%) Formative: Teamwork. Coach feedback using a rubric. Discussion after lectures. Feedback on involvement and input in the debate. Summative: (70%): a final team report and presentation of client based case that covers the analysis of the use-context, presents innovative solutions by applying context specific design methods. (30%): A reflective individual written exam on the role and position of designers in the context of low-resource settings.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: dfem-ide@tudelft.nl	

IDEM305	Designing Responsible AI	5
Course Coordinator	Dr. S. Colombo	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	Designing Responsible AI equips students to design AI solutions guided by ethical principles and an understanding of societal impact. By combining AI technology with ethics and speculative design, students gain the skills to navigate AI development conscientiously, ensuring they contribute to the advancement of responsible AI applications. The course is structured around three core activities e 1. Analysis of AI-related ethical issues: Students dissect real-world AI use cases in sectors like healthcare or social networks, focusing on ethical implications, biases, and privacy concerns. This segment develops their ability to critically and ethically analyze AI technologies. 2. Design futuring for AI systems: This creative module pushes students to envision future AI technologies. They conceptualize near-future AI applications through speculative design/design fiction, balancing technological advancements with societal needs and ethical considerations. 3. Critiquing AI designs: Students present AI design ideas, engaging in discussions, assessments and critiques. This fosters a culture of constructive criticism, essential for refining responsible AI systems. The course emphasizes hands-on AI experience, ethical reasoning, collaborative projects, and expert insights. Students learn to design AI systems that are socially responsible and ethically sound.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the most significant ethical issues related to AI and several design principles and frameworks for addressing them. 2. Analyze real-world AI applications in context using responsible design frameworks and methods (e.g., value-sensitive design, guidance ethics). 3. Envision responsible near-future AI systems by applying design-futuring methods (e.g., Ethnographic Experiential Futures). 4. Critically evaluate the ethical (de)merits of responsible AI system design visions.	
Education Method	In this course, students will participate in interactive lectures, design activities, and debates to learn about the ethics and responsible design of Artificial Intelligence. During self-study hours, they will read assigned papers and reflect on them. - Lectures will focus on the ethics of the design of technology, AI in particular, as well as methods for responsible design of AI, such as value-sensitive design and design futuring methods. - Design activities will revolve around a group project where a speculative design of a responsible AI system is created. - During critiques, students will learn how to argue for the (de)merits of peers design proposals in a principled and structured manner. - The papers will cover topics relevant to the theories and methods discussed and applied during contact hours. Students will also occasionally perform preparatory exercises to generate materials that will be used in subsequent contact hours.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5417 Artificial Intelligence and Society.	
Assessment	The attainment of learning objectives will be assessed through one group and one individual assignment. 1. Group assignment (60% of final grade; artifact + presentation; Week 7-8): Students will deliver an artifact that embodies the design vision (animation, video, or experiential prototype demo). The artifact is accompanied by a presentation that describes the rationale (argument) for the design proposal based on ethical analysis of target design situation. 2. Individual assignment (40% of final grade; short report; Week 10): Brief written assessment report on a peer groups design proposal (~750 words). Includes a reasoned selection of one of several assessment frameworks and a discussion of the pros and cons of the design proposal and its accompanying rationale. Formative assessment will include: Case analysis (group): Applying a responsible design framework or method (e.g., VSD) to analyze a provided design case and extrapolate values. Self-assessing the quality of the analysis through a checklist. Critique exercise (group): Engaging in a critique exercise aided by a guide. Peers provide feedback.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: drai-ide@tudelft.nl	

IDEM307	Generative AI and Design	5
Course Coordinator	Prof. G.W. Kortuem	
Contact Hours / Week x/x/x/x	4	
Education Period	1	
Start Education	1	
Course Contents	The course Generative AI and Design equips students to design and develop generative AI solutions and analyse their practical and ethical implications in a design context. Generative AI technologies such as Large Language Models (LLMs) and Latent Diffusion Models (LDMs), and generative AI tools such as ChatGPT and MidJourney, have the ability to generate content or data that is novel, creative, and often human-like. This course explores the impact of generative AI on design, and how we can use these new capabilities for creating new products and services, and new ways of designing.	
Study Goals	Upon finishing this course, students are able to: 1. Explain the working principles and ethical dimensions of generative AI technologies. 2. Prototype feasible and desirable generative AI products. 3. Analyse the implications of generative AI for design.	
Education Method	The course will use a combination of - Lectures and demonstrations. - Student exercises with demonstrations and presentations. - Reviews, critiques and discussions. - Assigned readings.	
Assessment	The study goals of this course will be assessed by: - Individual exam (50%) - Group work report (50%) Formative Assessment: - Weekly quizzes: students will self-assess their understanding using provided model answers. - Weekly exercises. Feedback will be provided during in-class reviews and discussions. Summative Assessment: - Individual exam (50%) focusing on understanding of working principles and ethical dimensions of generative AI. - Group project (50%) where students will document a functional prototype of a generative AI product, and analyse implications of generative AI for design.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: gaid-ide@tudelft.nl	

IDEM308	Graduation Launchpad	5
Course Coordinator	Dr. J. Wu	
Contact Hours / Week x/x/x/x	4	
Education Period	2 4	
Start Education	2 4	
Course Contents	The Graduation Launchpad prepares students for a graduation project related to research of the IDE faculty (https://www.tudelft.nl/en/ide/research). The Graduation Launchpad acts as a platform where students can interact with researchers to identify topics for graduation projects and learn how to approach, plan and execute a successful graduation project. After a general introduction of how to conduct high-quality graduation work, students will join a topic-specific group of researchers, potential project supervisors and students to conduct preliminary research and prepare their individual graduation project. At the end of the Graduation Launchpad, each student will have gained a firm grounding in a research topic, defined a feasible graduation project, identified a supervisor team and be ready to start a graduation project. Note: Taken the aim and set-up of this course into account, it is logical to take this course in the third semester/the quarter before you start the MSc Graduation Project. Therefore, the course is offered twice a year, in quarter 2 and quarter 4.	
Study Goals	Upon finishing this course, students are able to: 1. Reflect on personal research and design competencies. 2. Demonstrate advanced understanding of a design research topic as foundation for a graduation project. 3. Develop a credible and feasible graduation proposal.	
Education Method	The course uses a mix of education methods including lectures, exercises, and researcher-guided self-study . In the first part, students will be taught how to conduct high-quality graduation work using lectures and classroom exercises. In the second part, students will conduct individual research guided by small teams of researchers and prospective graduation project supervisors. Students are able to choose who to work with based on a published list of potential graduation topics. Each Delft Design Lab and IDE research group / lab will be offered to inject topics into the graduation launchpad and guide students. Teams of researchers / supervisors may elect their own style of guidance. Guidance will be tailored by researchers and prospective graduation project supervisors with respect to graduation topics.	
Assessment	The study goals of this course will be assessed by an individual assessment (100%) (pass/fail) Individual assessment based on a final graduation project proposal.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: gral-ide@tudelft.nl	

IDEM309	Repair & Recycling!	5
Course Coordinator	Dr.ir. S.F.J. Flipsen	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Repair and Recycle! addresses product lifetime extension and material recovery. RR! enables students to apply dedicated design approaches to improve the reparability and recyclability of products. The way in which products are designed determines the result of the repair process (fault diagnosis, dis and re-assembly, part replacement, testing) and the recycling process (collection, liberation, sorting, reprocessing). For improving on repair and recycling in practice, also the social, economic, and legislative aspects will to be considered.	
Study Goals	Upon finishing this course, students are able to: 1. Analyse the repairability of a product considering technical and design aspects. 2. Analyse the repairability of a product considering social, economic, regulatory aspects. 3. Analyse the recyclability of a product (social, economic, regulatory, technical and design aspects). 4. Redesign a product to improve ease of repair and ease of recycling. 5. Identify and discuss tensions between repairability, recyclability, and reliability of a product.	
Education Method	The course will offer 6 introductory lectures, self-study material and 8 short workshops (lecture 1 h + workshops 3 hours contact time per week). Students apply the knowledge by assessing and improving the reparability and recyclability of a specific product.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5422 Repair!	
Assessment	Team report (100%) on the redesign of a specific product including - Product specific assessment maps (disassembly, hotspot, recycling). - Social, economic, and regulatory aspects. - Redesign for repair and recycling. - Demo object and validation. - Evaluation of tensions.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: rr-ide@tudelft.nl	
Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

IDEM401	BlueDot: Design Contest	3
Course Coordinator	Ir. J.G. Steenkamer	
Contact Hours / Week x/x/x/x	4	
Education Period	2	
Start Education	2	
Course Contents	Design a lowtech product on a given theme from concept to a detailed prototype to fit in the BlueDot collection.	
Study Goals	Upon finishing this course, students are able to: 1. Provide insight into their personal potential as an industrial designer. 2. Enhance their design skills and experience. 3. Develope a market-ready product. 4. Enhance their presentation skills. 5. Enhance their model building skills. 6. Experience in working with real deadlines. 7. Build on their personal portfolio and resume.	
Education Method	Weekly meetings with a mentor according to the timetable.	
Prerequisites	You cannot take this course if you already completed the equivalent course ID5548 BlueDot: Design Contest.	
Assessment	The study goals of this course will be assessed by an assignment (100%) The project is assessed by the mentor. For this he/she will receive a concise report with a personal reflection on the course objectives, necessary information regarding the design and photos of the model. (Models must be presented to the supervisor before submitting them to the contest.)	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Contact: bluedot-ide@tudelft.nl	

IDEM402	BlueDot: Organisation	5
Course Coordinator	Ir. J.G. Steenkamer	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Course Contents	This project code is especially made for rewarding students who participated in the BlueDot Foundation as a member of the Contest Committee. Either they are paid through these ECTS points, OR they are given RAS maanden The Foundation BlueDot is a foundation for and by students with the objective to produce and market products designed by students. Every year the Foundation BlueDot has a new board and students may fulfill a role on that board. (Coursecode ID5552 / 6 ect) One of the activities that BlueDot develops each year is a Design Competition for students where the winners may - with he support of BlueDot - further develop and produce their design. The winning products are marketed by BlueDot. This contest is organised by the contest committee and students may fulfill a role on this organising committee. (Coursecode ID5552-17 / 3 ect) From BlueDots policy plan 2014/2015; Vision: The BlueDot Foundation strives to support all motivated students to develop their product design concepts into feasible and successful products. The Foundation thus creates a product portfolio, with which BlueDot can stand out as an innovative design label. The proceeds from sales are invested in new projects so that BlueDot can, financially independent, continue to support students. BlueDot Foundation aims to be a showcase of students creativity and diversity of Delft University of Technology. Mission: To offer students a unique learning-experience by supporting them in the development of their product design concepts into a market-ready product.	
Study Goals	Study goals for students on the board of the Foundation BlueDot and/or the contest committee: to gain experience with maintaining contact with producers and sellers to gain experience with the principles of entrepreneurship to gain experience with all facets of managing an organization to learn how to fulfill marketing and promotional tasks to learn how to fulfill selling tasks to guide a production process to learn about process management Furthermore students learn about conflict resolution, communication, problem-solving, leadership and evaluation. The drawing up and maintenance of a schedule designed to accomplish aims within the set time. The learning-experience provided by being a member of the board of the BlueDot Foundation proves to be useful during studies and is much valued by future employers. For students who are active within BlueDot it is also possible to fill in their board- or committee year in a more personal way. Examples are board members who educate themselves in product photography, programming a website, marketing research or legal matters.	
Education Method	A personal learning process by being a member of the board or through the organisation of an activity, for instance organising the BlueDot Design Competition.	
Assessment	A personal written evaluation of the board- or committee tasks completed, with a reflection on the course objectives described above.	
Enrolment / Application	Interested in becoming a BlueDot board- or committee member? Make an appointment with the BlueDot chairperson. website: www.stichtingbluedot.nl email: info@stichtingbluedot.nl For TUD-IDE students only.	

IDEM403	Research (3 ECTS)	3
Responsible Instructor	P. Wang	
Education Period	Different, to be announced	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	Students participate, on an individual basis, in one of the ongoing research projects in the Faculty IDE. The student gets acquainted with setting up and executing a research project. Important activities are reading relevant literature, formulating research questions, data collection, data analysis and writing report or scientific article.	
Study Goals	After a research project, students are able to participate in ongoing research in the faculty and are able to execute a research project.	
Education Method	The reseacrch project is executed and supervised in a small group or on an individual basis. The project will be formulated within ongoing research in the Faculty IDE. A researcher working on this specific research will supervise the student/ small group.	
Assessment	The research project is completed with a report or paper. The supervisor evaluates this report or paper.	
Enrolment / Application	For TUD IDE-students only. Contact professor Sicco Santema for more information: S.C.Santema@tudelft.nl	

IDEM404	Research (5 ECTS)	5
Responsible Instructor	P. Wang	
Education Period	Different, to be announced	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	Students participate, on an individual basis, in one of the ongoing research projects in the Faculty IDE. The student gets acquainted with setting up and executing a research project. Important activities are reading relevant literature, formulating research questions, data collection, data analysis and writing report or scientific article.	
Study Goals	After a research project, students are able to participate in ongoing research in the faculty and are able to execute a research project.	
Education Method	The reseacrch project is executed and supervised in a small group or on an individual basis. The project will be formulated within ongoing research in the Faculty IDE. A researcher working on this specific research will supervise the student/ small group.	
Assessment	The research project is completed with a report or paper. The supervisor evaluates this report or paper. After the research project is done and approved by the supervisor, the grade, the course ID and the student ID are send by email to s.c.santema@tudelft.nl, (Sicco Santema, the coordinator of the course). He will then put it in Osiris.	
Enrolment / Application	For TUD IDE-students only. Contact professor Sicco Santema for more information: S.C.Santema@tudelft.nl	

Year	**2024/2025**
Organization	**Industrial Design Engineering**
Education	**Master Integrated Product Design**

IPD Graduation Project (30 EC)

ID4190-16	Graduation Project (IPD)	30
Course Coordinator	Prof. G.W. Kortuem	
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Master’s Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: - Effectively collect, analyse, generate and evaluate knowledge required for the project - Justify his/her choices with respect to used methods and/or approaches used in the project. - Deliver a relevant (feasible, viable and desirable) project result - Effectively and thoroughly communicate to- and discuss with stakeholders involved in the project - Manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students’ work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; - A report providing insight in the process and project results, - A showcase; either a prototype or model, a movie or a poster - A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document 'Graduation Manual'. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

IDEM2000	Graduation Project IPD	30
Course Coordinator	Prof. G.W. Kortuem	
Education Period	None (Self Study)	
Start Education	1 2 3 4 5	
Exam Period	none	
Course Language	English	
Course Contents	This version of the graduation project is for students starting their master on or after September 2024. The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Masters Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: Effectively collect, analyse, generate and evaluate knowledge required for the project Justify his/her choices with respect to used methods and/or approaches used in the project. Deliver a relevant (feasible, viable and desirable) project result Effectively and thoroughly communicate to- and discuss with stakeholders involved in the project Manage a design/research project independently within the given time	
Assessment	During the graduation project, there are three formal moment when the students work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation projects is supported by means of a rubric. The deliverables for the graduation project consist of; A report providing insight in the process and project results, A showcase; either a prototype or model, a movie or a poster A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document Graduation Manual. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

J. Bourgeois

Unit	Industrieel Ontwerpen
Department	Internet of Things
Room	32.B-3-270

Dr. J.H. Boyle

Department	Emerging Materials

Prof.dr.ir. A. Bozzon

Department	Web Information Systems
Telephone	+31 15 27 87822
Room	32.B-3-370

Unit	Elektrotechn., Wisk. & Inform.
Department	Sustainable Design Engineering
Telephone	+31 15 27 87822
Room	32.B-3-370

S.C.M. Brand-de Groot

Unit	Industrieel Ontwerpen
Department	Form and Experience
Telephone	+31 15 27 82055
Room	32.C-1-570

Ir. M.J.J. Buijs

Department	DesIgning Value in Ecosystems

Dr. R.S.K. Chandrasegaran

Department	Creative Processes
Room	32.B-4-140

Dr. S. Colombo

Department	Human-Centred Artificial Intelligence

Prof.dr.ir. J.C. Diehl

Unit	Industrieel Ontwerpen
Department	Design for Sustainability
Telephone	+31 15 27 89729
Room	32.B-3-350

Dr.ir. E.L. Doubrovski

Unit	Industrieel Ontwerpen
Department	Mechatronic Design
Telephone	+31 15 27 86367
Room	32.B-3-020

Dr. R. van Egmond

Unit	Industrieel Ontwerpen
Department	Perceptual Intelligence
Telephone	+31 15 27 88324
Room	32.C-2-290

M. Filippi

Department	Form and Experience

Dr.ir. S.F.J. Flipsen

Unit	Industrieel Ontwerpen
Department	Design for Sustainability
Telephone	+31 15 27 89398
Room	32.B-3-060

Ir. M.A. Gielen

Unit	Industrieel Ontwerpen
Department	Codesigning Social Change
Telephone	+31 15 27 81744
Room	32.C-3-270

E.D. van Grondelle

Unit	Industrieel Ontwerpen
Department	Form and Experience
Telephone	+31 15 27 89488
Room	32.C-3-220

K.G. Heijne

Department	Creative Processes
Room	32.B-4-130

Ir. J.W. Hoftijzer

Unit	Industrieel Ontwerpen
Department	Human Factors
Telephone	+31 15 27 89325
Room	32.C-1-570

Dr. T. Huysmans

Unit	Industrieel Ontwerpen
Department	Applied Ergonomics and Design
Telephone	+31 15 27 81198

Prof.dr.ir. K.M.B. Jansen

Unit	Industrieel Ontwerpen
Department	Emerging Materials
Telephone	+31 15 27 86905
Room	32.B-3-170

Dr.ir. T.J. Jakiewicz

Department	Human Technology Relations

Dr. E. Karana

Unit	Industrieel Ontwerpen
Department	Emerging Materials
Telephone	+31 15 27 85726

Prof. G.W. Kortuem

Unit	Industrieel Ontwerpen
Department	Internet of Things
Telephone	+31 15 27 81952
Room	32.B-3-280

Dr. P.A. Lloyd

Department	Creative Processes
Telephone	+31 15 27 88334
Room	32.B-4-170

Dr. H.L. McQuillan

Department	Materials and Manufacturing

Dr. E. Niforatos

Department	Internet of Things
Room	32.B-3-390

Dr.ir. S.N. Paus-Buzink

Department	Human Factors

Ir. I.C. de Pauw

Unit	Industrieel Ontwerpen
Department	Design for Sustainability
Room	32.B-3-010

Prof.dr. S.C. Pont

Unit	Industrieel Ontwerpen
Department	Perceptual Intelligence
Telephone	+31 15 27 81791
Room	32.C-2-190

Dr.ing. M.C. Rozendaal

Unit	Industrieel Ontwerpen
Department	Human Technology Relations
Telephone	+31 15 27 86947
Room	32.C-3-250

Dr.ir. H.N.J. Schifferstein

Unit	Industrieel Ontwerpen
Department	Form and Experience
Telephone	+31 15 27 87896
Room	32.C-3-300

Dr. rer. nat. C. Schneegass

Department	Perceptual Intelligence

Dr. H.M.J.J. Snelders

Unit	Industrieel Ontwerpen
Department	Creative Processes
Telephone	+31 15 27 84644
Room	32.B-4-030

Dr. J. Soares de Oliveira Martins

Department	Mechatronic Design

Dr. Y. Song

Unit	Industrieel Ontwerpen
Department	Emerging Materials
Telephone	+31 15 27 88286
Room	32.B-3-220

B. Sprecher

Department	Design for Sustainability

Prof.dr. P.J. Stappers

Unit	Industrieel Ontwerpen
Department	Codesigning Social Change
Telephone	+31 15 27 85202
Room	32.C-3-130

Ir. J.G. Steenkamer

Department	Perceptual Intelligence

V. Tassinari

Department	Society, Culture and Critique

Dr. H. Verma

Department	Human-Centred Artificial Intelligence

Dr. V.T. Visch

Unit	Industrieel Ontwerpen
Department	Society, Culture and Critique
Telephone	+31 15 27 84951
Room	32.C-3-260

P. Wang

Department	Creative Processes

Dr. M.W.A. Wijntjes

Unit	Industrieel Ontwerpen
Department	Perceptual Intelligence
Telephone	+31 15 27 81835
Room	32.C-3-260

Dr. J. Wu

Unit	Industrieel Ontwerpen
Department	Materials and Manufacturing
Telephone	+31 15 27 84858